How criminal governance undermines elections*

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How criminal governance undermines elections

Abstract

Existing theories of criminal governance indicate that criminal groups can influence elections, sometimes acting as brokers. I argue that criminal groups' capacities as brokers stem from their control over territory and, subsequently, over voters. I theorize two criminal brokerage strategies, both made possible by territorial control: (1) *gatekeeping* limits rival candidates from entering the criminally governed area and (2) *corralling* influences vote choice. I test my theory in Rio de Janeiro, Brazil, using a natural experiment that leverages exogenous variation in voter assignment to voting booths, a novel dataset on criminal involvement in elections, and qualitative interviews. I show that fewer total candidates receive votes in areas with high gatekeeping, while criminally-aligned candidates are more likely to win in areas with high corralling. Both gatekeeping and corralling are associated with a decrease in electoral competition. These findings bring the literatures on clientelism and criminal governance closer together.

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In many democracies, criminal organizations play an important role in electoral politics. During campaigns, these criminal groups make deals, promises, and threats of violence, interacting with both candidates and voters. This paper illustrates *how* criminal groups deliver votes by operating as political brokers and how this affects democratic electoral competition.

A rich literature establishes that criminal groups are influential political actors (Barnes 2017; García-Ponce 2025; Skarbek 2024). Some studies acknowledge criminal groups' potential to mobilize voters, generating useful concepts that describe these relationships like "criminalized electoral politics" (Albarracín 2018), "double-barreled clientelism" (Arias 2006), or "armed clientelism" (Eaton 2006; Gallego 2018a). I build on this literature by elaborating two of these brokerage strategies in greater detail and empirically investigating their relationship to voting. In the clientelism literature, the role of criminal groups as brokers is still undertheorized, and they are often excluded from major typologies of brokers.[1] My theory brings these two literatures closer together by refining two mechanisms – *gatekeeping* and *corralling* – that explain how criminal groups can effectively deliver votes to their candidate partners.

I argue that criminal groups derive certain capacities from their control over territory that give them unprecedented social and militarized influence over voters, making them particularly effective as brokers. My theory, and the two mechanisms I refine, emerged from 18 months of fieldwork, including over 50 semi-structured interviews with candidates, staffers, voters, and community leaders in Rio de Janeiro, Brazil between the 2016 and 2020 election cycles. *Gatekeeping* obstructs rival candidates' access to voters. *Corralling*

1. Political brokers are "local intermediaries who provide targeted benefits and solve problems for their followers; in exchange, they request followers' participation... and often demand their votes" (Stokes, Dunning, Nazareno, and Brusco 2013).

influences residents' vote choice.[2] Both mechanisms stem from criminal governance.[3] I test the plausibility of each mechanism, leveraging new measures of these concepts and a novel empirical strategy.

To estimate the electoral returns of gatekeeping and corralling, I examine city council elections in Rio de Janeiro. I focus on local elections, which are easier for criminal groups to influence, according to evidence from Brazil (Gay 1993; Arias 2017), Italy (Alesina, Piccolo, and Pinotti 2019; Daniele and Dipoppa 2017), and Colombia (Gallego 2018b; García-Sánchez 2016; Gutiérrez Sanín 2015).

First, I create an original panel dataset from 2008-2020 about criminal involvement in local elections. I use the geolocated text from 1.45 million anonymous complaints to measure gatekeeping and corralling at the *favela*-level,[4] and combine this with fine-grained electoral data across four election cycles. Next, I leverage the exogenous assignment of residents to voting booths to compare voting between residents who live under criminal governance and their neighbors who do not, as if it were a natural experiment. I combine an original panel dataset on criminal governance with this data on criminal involvement in elections to analyze how gatekeeping and corralling vary by criminal group.

I show that election results in favelas where gatekeeping or corralling occur are less competitive than in favelas where they do not occur or in middle-class areas. Fewer unique candidates receive votes in favelas where criminal gatekeeping is present. Where criminal corralling is present, criminally-connected candidates are more likely to win that area's votes. I demonstrate that this is true for extortion racket *milícia*-affiliated candidates and to a lesser extent, for candidates affiliated with drug trafficking gangs. When gatekeeping, corralling, or both are present, elections are less competitive. Back-of-the-envelope calculations suggest

2. This term emerged inductively through my interview research. I discuss its connections to related concepts in the next section.

3. Criminal governance goes beyond physical presence; it is "the imposition of rules or restriction on behavior by a criminal organization" (Lessing 2021).

4. *Favelas* are informal settlements, often where criminal governance is concentrated.

that 24% of elected city councilors won because of votes mobilized from criminal brokers.

My findings make three main contributions. First, I synthesize and build on the extant scholarship to present a cogent, grounded framework about the consequences of criminal involvement in elections. Accounts of violence against or collusion with politicians theorize how criminal groups sway election outcomes (Albarracín 2018; Alesina et al. 2019; Arias 2006, 2017; Blume 2017; Córdova 2019; Daniele and Dipoppa 2017; Trejo and Ley 2020). Studies focused on the kaleidoscope of non-state actors in Colombia show that these armed groups intimidate rival politicians and coerce voters into supporting their favored candidate (Acemoglu, Robinson, and Santos 2013; Gutiérrez Sanín 2015; García-Sánchez 2016; Gallego 2018b; Steele 2011; Uribe 2025; Weintraub, Vargas, and Flores 2015). A related literature demonstrates that other actors (e.g., political parties or incumbents) use violence to dissuade rivals or manipulate vote choice (Birch, Daxecker, and Höglund 2020; Hafner-Burton, Hyde, and Jablonski 2014; Höglund 2009; LeBas 2013; Matanock and Staniland 2018; Siddiqui 2022, 2023; Turnbull 2021).

This paper combines extensive fieldwork with novel empirical tests to measure two modalities of criminal group intervention in elections. I test two mechanisms through which criminal brokers deliver votes (gatekeeping and corralling) that are salient in the Rio de Janeiro case. This paper moves the literature forward by empirically demonstrating that even subtle forms of influence, such as regulating candidates' access to voters, have the potential to influence electoral outcomes.

Second, my theory differentiates criminal groups from other classes of brokers by focusing on criminal groups' militarized and social control over territory. Theories of criminal governance have documented criminal groups behaving as brokers, delivering votes on a contractual basis (Arias 2006, 2017; Albarracín 2018; Eaton 2006; Gallego 2018a; Gutiérrez Sanín 2015, 2019). Despite this, theories of clientelism often omit criminal organizations in their rich typologies of brokers (Holland and Palmer-Rubin 2015; Novaes 2018; Stokes

et al. 2013). I advance both literatures by arguing that criminal groups' ability to be effective brokers is related to their ability to control territory and the population. By detailing how criminal territorial control matters, this paper sets the stage for future research on this crucial yet underexplored category of political broker.

Third, my results help to explain why politicians may tolerate the risks of engaging with criminal groups. While tough-on-crime platforms remain popular at the polls (Flores-Macías and Zarkin 2022; Holland 2013), my findings suggest that criminal groups can help candidates win. When candidates hire criminal brokers, it creates an agency dilemma in which politicians may want to weaken organized crime broadly, but not necessarily in the neighborhoods that vote for them. This tension makes it difficult to fight crime effectively and get crime out of politics. This paper provides a theory that illuminates what politicians can gain from partnering with criminal groups, adding to the recent body of work about the state's complex incentives to address the root cause of organized crime (Blattman 2025; Lessing 2017; Trejo and Ley 2020). Overall, the findings from this paper speak to open questions about criminal influence over politics, and how certain politicians leverage criminal governance for electoral gain.

Criminal groups as brokers

When criminal groups govern, they differ from other brokers because they have a certain degree of militarized control over territory. My theory begins from the premise that criminal groups have medium to high levels of consolidated governance, meaning that they already are a local authority and have existing power and relationships in their community (e.g., the "collaborative" or "divided" criminal regimes in Arias (2017)).

Criminal groups of this type often exert broad control over social life in the community. They build hierarchical ties with voters and fully or partially control resource distribution

networks, sometimes providing goods and services as if they were the state (Barnes 2025; García-Sánchez 2016; Magaloni, Franco-Vivanco, and Melo 2020). Often, other local actors defer to the criminal group (Arias 2006; Gay 1993; LeBas 2013; Moncada 2013). These patterns are similar to the brokerage dynamics between patrimonial leaders and their constituents. Distributive politics scholars argue that such linkages are the "social cement" for particularly durable clientelistic connections (Baldwin 2015; Koter 2013; Kramon 2019; Stokes 2009).

However, unlike patrimonial elites, certain criminal groups have a degree of militarized *territorial control* that they can use to influence state activity and limit rival armed group activity. Criminal groups may have such control even when they do not aspire to "full" territorial control vis-à-vis the state (e.g., state absence) (Barnes 2022). The building blocks of this argument focus on criminal groups with such control.[5] First, influencing state activity encompasses a range of behaviors, from the prevention to facilitation of state delivery of public services, justice, or public security. Second, criminal groups have the militarized control to prevent or limit entry by rival armed groups. While criminal groups often pursue territorial control to prepare for or prevent violent conflict with rivals (Barnes 2022; Rios 2013), they can leverage it for other reasons, including influencing elections (Gallego 2018b; García-Sánchez 2016).

The coercive power that comes with militarized territorial control is unavailable to other brokers. Even other types of brokers that use violence – such as certain ethnic leaders – do not have this degree of coercive power. Extending the argument in Mares and Young (2016), who emphasize criminal groups' use of violence to obtain votes, I posit that criminal groups' efficacy as a broker stems less from their *use* of violence than from their control over territory, which *enables* them to use a broad range of brokerage strategies, including but not

5. Many criminal groups have little to no territorial control. I address the theory's relevance to them at the end of this section.

limited to violence. For example, a community organizer in Rio de Janeiro described how criminal groups leverage territorial control to supersede others' electoral preferences: "The pastor's, the personal trainer's, the schoolteacher's... everyone else's candidates have to get approved by the criminal organization before they can campaign here" (Interview 1).[6]

The level of militarized control over territory enables criminal groups to govern in two ways that are relevant to the study of brokerage. First, it enables them to have *border control*, or control over the perimeter of the governed area. Border control is important for brokerage because it serves a regulatory purpose, discouraging certain actors from entering their territory. For example, criminal groups commonly employ "lookouts" and "guards" to monitor urban turf, permitting access for friends and denying it for foes (Dowdney 2003). Second, it enables them to have *internal control*, or control over residential life, such as, curfews, taxation, or dispute resolution (Lessing 2021).[7] Internal control is important for brokerage because it characterizes how the criminal group influences the population. Some groups obtain internal control through coercion and fear (Magaloni, Robles, Matanock, Díaz-Cayeros, and Romero 2020). Others accrue it through social embeddedness (Blume 2021; Dowdney 2003).[8]

I argue that criminal groups' militarized capacity to regulate access to the community and influence residents, stemming from border and internal control, respectively, sets them apart from competing political brokers. Unlike other independent brokers in their communities (e.g., religious or civic leaders (Holland and Palmer-Rubin 2015)), only criminal groups can leverage militarized border control to regulate which candidates may enter and campaign. Additionally, few other types of brokers have such a broad range of internal governing tools to

6. See the online Supplemental Information (SI) Section G (p. 25) for a discussion of reporting and ethics for all qualitative sources.

7. I use the term *internal control* to differentiate it from the related concept of armed group *social control*, which forms during civil war (Jentzsch and Steele 2023).

8. Social embeddedness is not a substitute for coercive capacity, because criminal groups that refrain from using coercion may still access it as a last resort (Arias 2006; Glenny 2015).

precisely target vote buying or deliver threats, and those that do (e.g., patrimonial leaders) usually lack militarized border control. Together, criminal brokers' border and internal control differentiate them from other contemporary brokers.[9]

Where criminal organizations have militarized control over territory, standard explanations of brokerage relationships may no longer be sufficient to describe the electoral dynamics among candidates, criminal groups, and voters. I suggest two mechanisms through which criminal groups can deliver votes: *gatekeeping* and *corralling*.

Candidate gatekeeping

One way criminal groups affect electoral outcomes is by gatekeeping, or restricting the candidate pool. Criminal groups leverage border control to selectively allow some candidates to physically enter the community, and others not. This shapes the information that voters have about the candidate pool, limiting contestation. The literature on armed groups and elections documents this behavior in Colombia (Gallego 2018b; García-Sánchez 2016) and Brazil (Albarracín 2018; Arias 2017). I formalize its logic below.

Criminal brokers can shape information about candidates through two channels: facilitating their preferred candidate's access to voters, or obstructing rival candidates' access to voters. The first channel draws on informational theories of clientelism, in which brokers signal candidate viability by increasing exposure to the candidate (Muñoz Chirinos 2019). Brokers can intensify this physical contact through in-person events or advertising (Kramon 2016). The second channel – preventing rival candidates from accessing voters – follows from their border control. Criminal groups can grant *their* candidate entry and obstruct access to the rest. When rivals cannot physically enter the community, I expect this to obstruct their efforts to advertise. Voter complaints[10] support this logic; one states:

9. Criminal groups that are part of a political party are not the focus of this paper, but have also been documented mobilizing voters (Siddiqui 2022).
10. See the data section for more details about the complaints.

> "Cars with other candidates' decals are prohibited from moving about FAVELA. CANDIDATES A and B struck a deal with the traffickers, buying the entire community, prohibiting posters, decals, pamphlets, and even activity involving other candidates on residents' social media" (Complaint 1).

When gatekeeping occurs, voters should receive a disproportionate amount of information about the criminal group's candidate and little about rivals. My interview evidence shows that excluded candidates report being unable to campaign without bargains with criminal leaders. Twelve candidates recounted being told they could not campaign somewhere, or were asked to leave favelas where they lacked such arrangements. Candidates and staffers acknowledge that gatekeeping increases the chances of winning. Four used an idiom that means "closing a deal,"[11] signaling mutual understanding about the exclusivity that gatekeeping confers. Many who hired criminal brokers expected to be the only ones campaigning in the area.

The electoral returns to gatekeeping may be particularly salient in cognitively burdensome elections with many candidates. In such races, few candidates receive press, so in-person strategies are essential (Gay 1993). Candidates who are gatekept out have limited options to shape voter information from afar.

When criminal groups gatekeep, I hypothesize that they will reduce the effective candidate pool in their area. Gatekeeping limits contestation through an informational channel, by privileging information about some candidates and obstructing information about others.

Voter corralling

Another way in which criminal brokers affect electoral outcomes is by corralling, or influencing vote choice. My conceptualization, drawn from my interview evidence, builds on decades

11. In Portuguese, *fechar com (alguem)*.

of research on rural clientelism in Brazil, beginning with the term *curral eleitoral* (electoral corral) that broadly describes the patron-client ties linking oligarchs and voters in studies of *coronelismo* (Hagopian 1996; Leal 1949).[12] Clientelism is also an urban phenomenon; scholars have noted its ubiquity in Brazilian cities since before the military dictatorship (Fischer 2008; Leeds and Leeds 1978). Today, the terms *curral eleitoral* and *coronelismo* are associated with various concepts about favela clientelism (Arias 2006, 2017; Gay 1993, 1999; Leeds 1996; McCann 2014).

I extend this literature by offering a clearer conceptualization of the term corralling and explaining how it is used by criminal groups. Unlike the broader concept of *coronelismo*, corralling specifically refers to the ways that criminal brokers interact with the population to mobilize their vote. Corralling is predicated on internal control: when criminal groups corral, they leverage their control over residential life to sway vote choice, using persuasive or coercive strategies.

Criminal brokers use internal control to persuade voters through the provision of short- or long-term rewards (Gans-Morse, Mazzuca, and Nichter 2014; Stokes et al. 2013), ranging from cash payments to social and health services, and even to small plots of land in the communities they govern. Criminal brokers might also genuinely persuade voters when they have high social capital or are embedded in the territory (Auerbach and Thachil 2018; Zarazaga 2015).

Criminal groups also use internal control to coerce voters, drawing from both their control over residential life and their coercive capacity. The severity of threats varies widely, from terrorizing and even killing voters (e.g., Colombia's paramilitary group tactics in Acemoglu et al. (2013), Steele (2011), and Uribe (2025)) to minimal violence or verbal threats (e.g., Albarracín's (2018) "delegation" type). These threats appear effective at the group

12. During the Old Republic (1889-1930), the *coronel* "shepherded his followers to the polls on the backs of farm trucks," generating agricultural metaphors ("herd vote" and "electoral corral") that describe the level of physical and social control (Hagopian 1996).

level, likely because of internal control: threatening a few voters can inspire fear across the community.

In classic theories of clientelism, monitoring is key to influencing vote choice. My interviews suggest that the monitoring that accompanies criminal corralling operates at the group-level: criminal groups sell the votes "wholesale" of the entire community to their candidate partners.[13] Even in the absence of individual-level monitoring, one candidate claimed, "you can count on having those votes" (Interview 3) delivered by a criminal group, but the same was not true for other clientelistic strategies. One staffer admitted that coercive corralling yielded the most votes vis-à-vis other strategies (Interview 4). Existing studies of *milícias* in Rio de Janeiro also note group-level monitoring behaviors (Cano and Duarte 2012). Individual monitoring is unusual in a case like Brazil with high ballot integrity, and appears unnecessary for the criminal broker, given that group-level threats seem effective.[14] One voter states, "CANDIDATE, in addition to offering money to buy votes, threatens residents that if he discovers that they did not vote, they will suffer the consequences. CANDIDATE is involved with the *milícia*" (Complaint 2).

When criminal groups corral, I hypothesize that they will influence vote choice through persuasion, coercion, or both, even in the absence of individual-level monitoring.

Gatekeeping and corralling reduce electoral competition

I argue that gatekeeping, corralling, or their joint use reduces electoral competition. They are not mutually exclusive and can be complementary. As gatekeeping reduces the number of candidates that can access voters, it facilitates corralling for the criminal group's preferred candidate. Voters confirm their joint usage:

13. "Wholesale" agreements have been documented elsewhere when brokers enforce clientelistic bargains at the group level (Kramon 2019; Rueda 2017).

14. Individual monitoring, such as criminal groups demanding ballot cell phone photos, appears infrequently in the voter complaints.

> "Today when I went to vote, I was coerced by *milícias* to vote for CANDIDATE. There were lots of flyers and stickers for him surrounding the polling station entrance, scattered across the floor. He was the only one who campaigned in my neighborhood" (Complaint 3).

These two strategies can be used together but not always are. This could be due to different opportunity costs of engaging in either activity, given available resources. I test for this possibility by comparing criminal groups with different resource allocations.

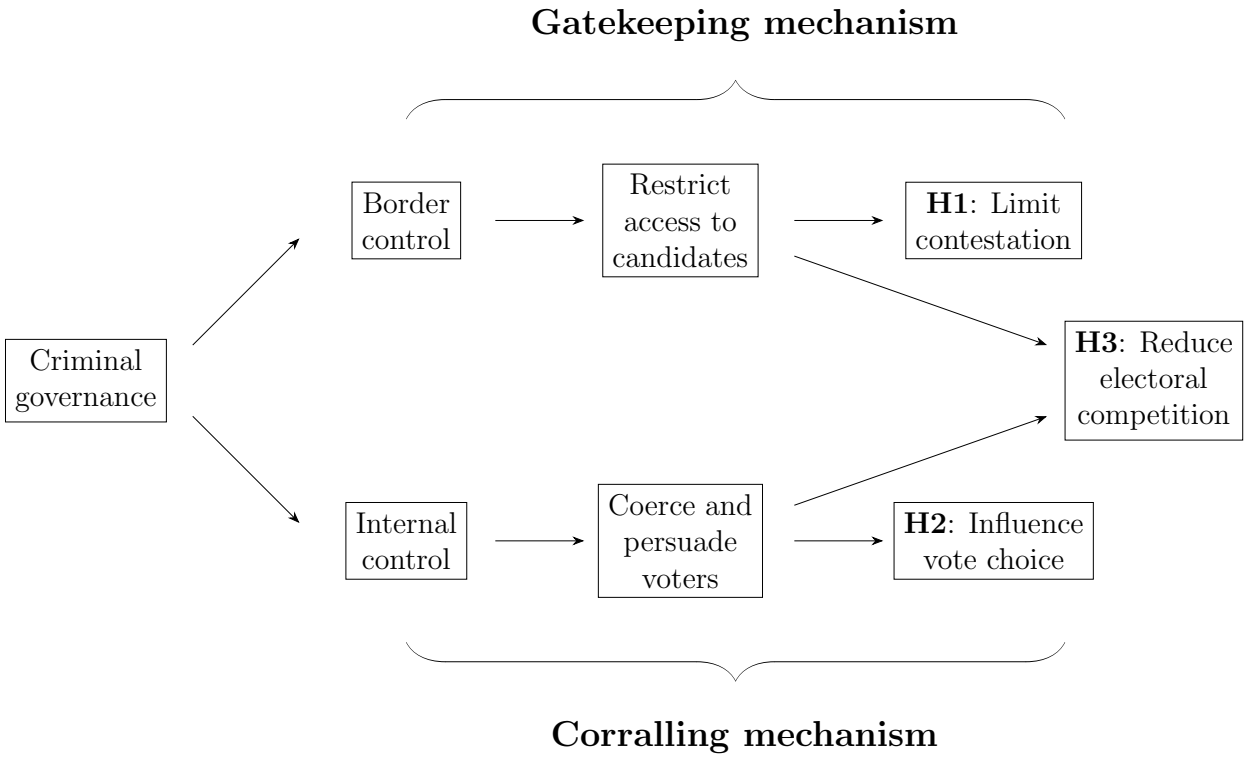
To summarize, gatekeeping and corralling are brokerage strategies that are derived from two components of criminal governance, border and internal control. Militarized territorial control enables criminal brokers to effectively practice border and internal control, which enables gatekeeping and corralling, respectively. What about criminal groups with little to no militarized control or a light territorial presence? It is unlikely that such groups will be able to gatekeep or coercively corral to the same degree. But if these groups have social capital, they may be able to persuade voters through interactions that are not as linked to territory, following the "social control" logic in Jentzsch and Steele (2023).

In contrast, for groups with militarized territorial control, I expect that a higher degree of control will make them better brokers. The effectiveness of gatekeeping and corralling should increase as they consolidate border and internal control, respectively.[15] Criminal groups with greater control over the perimeter will be better at obstructing entry, and those with greater control over residential life better at coercion or persuasion. One does not necessarily require a higher level of territorial control than the other, because gatekeeping and corralling correspond to different governance problems.

Figure 1 summarizes my argument. This argument rests on four scope conditions: (1) the actors studied are territorial criminal groups; (2) criminally governed areas are controlled

15. Though I am unable to directly test these claims, this insight aligns with the literature arguing that criminal groups with consolidated control can more effectively pursue governance, security, and political goals (Arias 2013; Lessing 2021).

Figure 1: Structure of the argument



by a single group, although I explore the theory's applicability to contested territories in the robustness checks; (3) criminal governance is consolidated; and (4) candidate-criminal group deals are broadly defined, encompassing both longstanding deals[16] and short-term opportunistic deals.[17] The terms of criminal brokerage are not in the scope of this paper since *gatekeeping* and *corralling* can be observed across many arrangements.

Elections and organized crime in Rio de Janeiro

This paper focuses on municipal legislative elections, because a criminal group's support may more easily tip the scale in such contests than in higher-level races. Brazil's electoral rules

16. E.g., Sicilian Mafia's alliance with the Christian Democrats (Gambetta 1996).
17. E.g., Indonesian parties and the protection rackets for hire (Tajima 2018).

facilitate intense competition for votes.[18] In the 2016 Rio de Janeiro city council election, 1,628 candidates affiliated with 35 different parties competed for 51 seats. Dozens of marginal candidates won or lost by a few thousand votes.

Estimates suggest that criminal groups govern between one quarter and one half of the city's 6.74 million inhabitants (Satriano 2020). These voters live in communities of various sizes, primarily *favelas*. Favelas are low-income settlements characterized by high population density and varying degrees of social exclusion (Perlman 2010). There are hundreds of favelas scattered throughout the city; they are often contrasted with the surrounding neighborhoods in the formal sector, colloquially "the asphalt." The government neglected residents in favelas long before the birth of today's criminal groups (Fischer 2008; Misse 1999; Zaluar 1994), but these inequalities have widened over decades. Today, most of the city's favelas are governed by organized crime.

Two main categories of criminal groups govern favelas. First, drug trafficking organizations use the favelas they control for business purposes, including storing, transporting, or selling drugs (Arias 2006; Barnes 2025; Lessing 2017). Each favela is governed by one of three major syndicates[19] but operates as a decentralized member of a network, like a franchise. The second type of criminal group is vigilante-style extortion rackets, called *milícias* (Cano and Duarte 2012). The milícias – often associated with law enforcement – extort residents and businesses for "security" payments and control access to utilities, transportation, and loans. Milícias are also decentralized; milícia leaders control neighborhood-level protection rackets and coercive power. The decentralized nature of criminal governance means that there are hundreds of "criminal kingdoms" across the city (Arias 2006). Each favela's local leader – trafficker or milícia – has significant autonomy in setting the rules in their territory;

18. Legislative elections in Brazil use an open-list proportional representation system in multimember districts. All candidates are elected at-large, although many establish informal strongholds (Ames 2001). Parties may each nominate up to 1.5 times the number of seats.

19. These are ADA (*Amigos dos Amigos*), CV (*Comando Vermelho*), and TCP (*Terceiro Comando Puro*).

these governing styles may even vary between favelas governed by the same group (Magaloni, Franco-Vivanco, et al. 2020).

This decentralization also applies to favela electoral politics: if a candidate pays a criminal broker in one favela, this does not purchase access to all favelas dominated by that group. Local criminal bosses can thus charge a premium for gatekeeping or corralling services, profiting from the decentralized market structure. Candidates are aware that the price of entry doesn't extend far beyond the favela's perimeter. One candidate described paying the gatekeeping fee to safely campaign in a favela, but noted that he did not expect to gain votes in other locations where he did not purchase access (Interview 5).

Criminal-politician ties have made the news in Rio de Janeiro. Famously, there was a state-level congressional investigation into certain milícia groups for their connections to the state, including electoral politics. This investigation, the Parliamentary Inquiry Commission on Milícias (*CPI das Milícias)* occurred in 2008 and led to the high-profile removal of several local- and state-level politicians (ALERJ 2008). The extensive *CPI das Milícias* report, available online, details how milícias practiced gatekeeping and corralling for their candidates' electoral gain, among other things.

More recently, journalists have documented criminal gatekeeping and corralling in both milícia and drug trafficking favelas. They note that criminal brokers often set a flat rate for gatekeeping rather than charging per vote (Araújo and Otávio 2018), which typically includes exclusive or near-exclusive access to the favela. Many criminal leaders offer additional services to corral residents, such as hosting events or door-to-door canvassing.

Research design

If my theory is correct, when criminal gatekeeping and criminal corralling occur, I expect to observe the corresponding changes in candidate entry, vote choice, and electoral competition.

This section presents the data and empirical strategy.

Data

Criminal gatekeeping and corralling

I construct an original panel database of criminal gatekeeping and corralling in favelas. I use data from *Disque Denúncia*, a non-governmental organization that operates an anonymous 24-hour tip line. They are independent from the police but pass along time-sensitive, life-threatening calls to the police, while protecting callers' anonymity. The anonymity offers callers protection, which encourages reporting.[20] I obtained text summaries and the reported location of each call for approximately 1.45 million calls from 2008 to 2020 within the city of Rio de Janeiro.

I measure the prevalence of gatekeeping and corralling in the call summaries in five steps, detailed in SI Section A (p. 2). First, I geocode all calls and classify them as inside or outside a favela. Second, I restrict the sample to the campaign season, resulting in 42,889 calls inside favelas. Third, I create word lists (SI Table A1, p. 5) for four topics: (1) organized crime, (2) elections, in general, (3) gatekeeping, and (4) corralling. For each call, I code four binary variables indicating whether each topic is mentioned. Fourth, I summarize each topic by favela-year by generating a binary and count variable: the binary variable takes a value of 1 if any call mentions the topic in a given election cycle, and the count variable tallies the calls. Summary statistics are shown in Table 1. The third column reports the within-polling-station-year standard deviations, SD(w), which match the fixed effects in Equation 1 (below). There is considerable within-variation in the independent variables even after controlling for fixed effects.[21]

20. Magaloni, Franco-Vivanco, et al. (2020) have validated this data to study organized crime in Rio de Janeiro.

21. This is consistent with best practices for reporting fixed effects estimators (Mummolo and Peterson 2018).

Fifth, I construct the criminal gatekeeping and criminal corralling binary and count variables by labeling calls where organized crime co-occurs with either gatekeeping or corralling. The favela-year binary variable reported in Table 1 takes a value of 1 if any call mentions both organized crime and corralling or gatekeeping, and the count variable sums the number of calls. I read a random sample of calls to ensure construct validity. For example, criminal gatekeeping calls mention "drug traffickers... removing signs and other forms of campaign propaganda from candidates who are not their preferred choices" and criminal corralling calls describe "two individuals... believed to be milícia members, are distributing campaign pamphlets to voters" (see examples in SI Figures A1 and A2 on pages 3-4).

Electoral outcomes

To calculate electoral outcomes, I create an original dataset drawing from voting-booth-level returns from the Supreme Electoral Court (TSE). The voting booth is the smallest unit at which results are reported; approximately 300-500 people vote per booth. Table 1 also reports the summary statistics for the main dependent variables, and their distributions are shown in SI Figure F1 (p. 17).

The first outcome focuses on the observable implications of candidate gatekeeping, measuring whether criminal groups are reducing the effective candidate pool in favelas. I operationalize this by summing the number of candidates that receive at least one vote per booth. Given that there are more candidates in the race than voters per booth, it is statistically impossible for all candidates to receive a vote.

The second, third, and fourth outcomes are observable implications of voter corralling (detailed in SI Section B, pp. 6-8.). First, I measure whether milícia-affiliated candidates receive the most votes per voting booth. I collected a list of 18 names of Rio de Janeiro city council candidates mentioned in the unredacted *CPI das Milícias* report (see the previous section), which I then link to their electoral returns across election years. "Vote choice

Table 1: Summary statistics

Topic	Mean	SD	SD(w)	Min	Max
Calls about organized crime topics					
Organized crime (binary)	0.669	0.471	0.467	0	1
Organized crime (count)	15.992	33.803	32.650	0	337
Calls about election topics					
Elections, in general (binary)	0.118	0.322	0.317	0	1
Elections, in general (count)	0.657	2.792	2.745	0	26
Gatekeeping (binary)	0.058	0.234	0.230	0	1
Gatekeeping (count)	0.263	1.370	1.352	0	14
Corralling (binary)	0.037	0.188	0.186	0	1
Corralling (count)	0.111	0.723	0.715	0	8
Calls about organized crime and election topics					
Criminal gatekeeping (binary)	0.050	0.218	0.214	0	1
Criminal gatekeeping (count)	0.245	1.357	1.338	0	14
Criminal corralling (binary)	0.027	0.162	0.160	0	1
Criminal corralling (count)	0.093	0.680	0.672	0	8
Electoral outcomes					
Gatekeeping: N candidates receiving votes	108.758	17.905	17.580	22	192
Corralling: Vote choice (milícia)	0.073	0.260	0.255	0	1
Corralling: Vote choice (all)	0.174	0.379	0.365	0	1
Corralling: Vote choice (trafficking)	0.092	0.289	0.283	0	1
HHI (log)	−3.146	0.395	0.387	-4.091	-0.896
Observations	29,026				

Note: Unit of analysis is the ballot-box-year. SD(w) refers to the within-polling-place-year standard deviation.

(milícia)" is a binary indicator that measures whether an (alleged) milícia-affiliated candidate received the most votes at a voting booth.

The third and fourth outcomes take a similar approach. I hand-coded each candidate mentioned in the *Disque Denúncia* calls to generate a list of 35 candidates: all 18 candidates in the *CPI* report and 17 additional candidates that have alleged relationships with milícias or drug trafficking gangs. I create a binary indicator ("Vote choice (all)") that measures whether any of these 35 candidates received the most votes at a voting booth. "Vote choice (traf.)" then narrows the indicator to only those candidates mentioned in *Disque Denúncia* for electoral affiliations with trafficking groups, and measures whether they receive the most votes per booth.

The benefits of measuring corralling using the first measure are that it is sourced from a public document which has been widely used in social science research (Hidalgo and Lessing 2019; Paes Manso 2020; Pantaleão and Montini 2025). However, it only measures supposed milícia ties in 2008. The benefits of using *Disque Denúncia* measures are that they include candidates with ties to both group types and candidates mentioned more recently. The disadvantages of the *Disque Denúncia* measures are that the names cannot be disclosed for data confidentiality restrictions. Though neither measure is without flaws, measuring criminal influence over vote choice is a difficult construct to capture, and together, these three variables illuminate various aspects of such influence.

The fifth outcome is a summary indicator of electoral competition, resulting from gatekeeping, corralling, or both. I use the Herfindahl-Hirschman Index (HHI), a standard measure of monopoly concentration (see SI Section B.2, p. 8, for the equation). A larger value of the HHI means that one candidate receives a disproportionately high share of the votes per booth. For all specifications, I take the log of this variable to reduce skew.

Together, these outcomes paint a complete picture of the relationship between criminal governance and voting. If they move as predicted, they demonstrate that the electoral race

is growing less competitive as votes are being redirected from low vote-earners to criminally-connected candidates.

Control variables

I address concerns about omitted variable bias by including observable controls that might confound the relationship between gatekeeping or corralling and electoral outcomes, detailed in SI Section C (pp. 8-13). First, I control for the dominant criminal group(s) in the favela for each election-year, to address concerns that gatekeeping or corralling is only practiced by certain criminal group types. To generate this variable, I construct an original dataset at the favela-year level that measures the governing group, using the same *Disque Denúncia* complaints. I include indicators for the **CV**, **ADA**, **TCP**, and **Milícia** criminal groups, and an additional "**Criminal conflict**" indicator variable if the favela has an active conflict during the election. I condition secondary analyses on group type throughout the results section.

Second, I add other favela-level controls: 1) population, since organized crime might be stronger in larger favelas; 2) the distance to the polling station of interest, because criminal groups might have more influence nearby; and 3) favela income per capita, since poverty is an important predictor of clientelism. Third, I control for ballot-box-level voter characteristics to address potentially confounding variables that can influence participation or susceptibility to clientelism, including age, education level, gender, and civil status.

Empirical strategy

I use as-if random evidence to estimate the association between criminal gatekeeping or criminal corralling and voting. This paper exploits exogenous variation in voting booth assignment to demonstrate that favela residents' voting behavior differs from their neighbors' on "the asphalt," and that these differences are especially pronounced in favelas where

criminal gatekeeping and corralling occur.

A key assumption underpinning this paper's empirical strategy is that criminal governance is restricted to Rio de Janeiro's favelas. While not necessarily a feature of *all* favelas, criminal groups govern most favelas, and their presence is most obvious in favelas. My original data indicates that 57-70% of favelas are criminally governed, comprising 84-89% of favela residents, meaning that all large and medium-sized favelas are criminally governed (SI Table C2, p. 11). Criminal control extends to, but rarely beyond, the favela boundary.[22] Decades of qualitative work has documented different voting patterns between favela and non-favela ("the asphalt") residents (Gay 1993, 1999; Holston 2009; Perlman 2010).[23]

I approximate random assignment to the criminal gatekeeping or criminal corralling "treatment" by leveraging Brazil's exogenous voting booth assignment procedure, explained in more detail in SI Section D (pp. 13-15) and summarized below. First, when voters register, they choose the polling station within their electoral district where they want to vote.[24] Most choose polling stations close to their homes. Then, voters are randomly assigned to one of many voting booths *within their selected* polling station through a federal-level electronic assignment procedure. The software aims for an equal number of voters across booths in a polling station, and randomly assigns new registrants accordingly.[25] New voting booths are created if existing ones are at capacity. Voting booth assignment carries over year-to-year unless voters change their registration or become inactive.

22. Residents living under criminal control outside favelas represent a small share of the criminally-governed population vis-à-vis favela residents. Marinatto and Soares (2015) estimate that the size of the population of criminally controlled public housing residents — one major area outside favelas where criminal groups might govern — comprises less than 3% of the criminally governed population in Rio de Janeiro.

23. The *Disque Denúncia* calls support this assumption. Few election-related calls located on the asphalt are about organized crime, and 96% of them were located within 500 meters of a favela.

24. Electoral districts, *zonas eleitorais*, are federal reporting units encompassing many polling stations.

25. TSE Resolution N. 23.544, Articles 41, 44, and 66.

The implications of this assignment process are that voters who live in the same favela vote at many nearby locations, and intermingle with asphalt residents at the voting booth. Importantly, this also means that different voting booths – in the same polling place – will have different proportions of voters that reside in favelas. This imbalance is exogenously determined. Generally, polling stations are not located inside favelas due to accessibility and building code standards, so favela residents must leave their communities to vote.

Favela voters comprise the treatment group. I use new data, obtained from an information request, to look *within* the voting booth and measure the share of voters who live in a favela. For every voting booth, I obtained a number between 0 and 1, indicating the proportion of favela residents assigned to vote there. I leverage the exogenous imbalance that results from as-if-random voting booth assignment, which allows me to precisely quantify how favela resident presence in a voting booth affects voting, as if it were a natural experiment. This design solves the ecological inference problem by leveraging the as-if-random fraction of favela voters casting ballots in each voting booth, yielding the most precise known estimates of favela voting behavior. Equation 1 shows the model used to leverage this exogenous imbalance in voter assignment:

$$Y_{it} = \alpha + \gamma(DD_{it} \times Share_favela_i) + \eta\mathbf{X} + \pi_p + \phi_t + \epsilon_{i,p} \tag{1}$$

In this equation, i and t correspond to each voting booth and election year, respectively, and Y_{it} represents one of the four electoral outcomes. $Share_favela_i$ is the share of voters at each booth who live in its most represented favela, and is logged to reduce skew. The main independent variable, DD_{it}, is an indicator that takes on a value of 1 if there are *Disque Denúncia* reports of criminal gatekeeping or criminal corralling from the favela with the highest share of voters at the voting booth. I also estimate models where DD_{it} is a count variable. γ is the coefficient of interest. I include a vector of favela- and ballot-box-

level control variables ($\mathbf{X}$), polling station (π_p) and election-year intercepts (ϕ_t), and cluster standard errors at the level of randomization, the polling-station-level (p). The fixed effects control for unobserved, time-invariant characteristics at the polling station level, and for time-specific shocks from election to election.

Results

Table 2 summarizes the mechanism-specific results, reporting the interactive effects of favela residency with criminal gatekeeping or criminal corralling on voting. Binary and count independent variables are used in the odd and even models, respectively. The unit of analysis in all models is the voting booth, pooled across the 2008, 2012, 2016, and 2020 elections.

Column 1 shows that fewer candidates receive votes when criminal gatekeeping occurs at the local favela, when favela residents comprise a larger share of the booth's voters, and when these factors interact. On average, 9.8 fewer candidates receive votes when criminal gatekeeping is present, and this further decreases as the share of favela residents at the voting booth increases. All else equal, in the presence of criminal gatekeeping, 1.8 fewer candidates receive votes in booths with a high (75%) versus low (25%) share of favela voters.

Column 2 suggests that gatekeeping also operates on the intensive margin. A one standard deviation increase in the residualized interaction of *Share favela* and gatekeeping (SD = 3.55) corresponds to 1.64 fewer candidates receiving votes.[26] Though these are small increases vis-à-vis the outcome's mean, they are substantively meaningful, given that the fixed effects alone explain 55% of the variation in the outcome. These findings support the argument that criminal groups restrict candidate entry by gatekeeping, and that fewer candidates receive votes as gatekeeping increases. When holding the share of favela residents constant across booths, the presence of gatekeeping is systematically associated with

26. I residualize the treatment with respect to polling station and election-year fixed effects, as in Mummolo and Peterson (2018).

Table 2: Criminal gatekeeping and corralling on voting

	H1: Gatekeeping		**H2: Corralling**			
	N candidates		Vote choice (milícia)		Vote choice (all)	
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	-1.927^{**}	-0.461^{**}				
	(0.447)	(0.093)				
Criminal gatekeeping	-7.872^{**}	-1.950^{**}				
	(1.697)	(0.338)				
Criminal corralling × Share favela			0.066^{**}	0.015^{**}	0.068^{**}	0.026^{**}
			(0.020)	(0.005)	(0.017)	(0.006)
Criminal corralling			0.230^{**}	0.037^{*}	0.496^{**}	0.138^{**}
			(0.072)	(0.017)	(0.076)	(0.016)
Share favela	-1.622^{**}	-1.612^{**}	-0.006	-0.005	-0.008	-0.008
	(0.302)	(0.302)	(0.006)	(0.006)	(0.009)	(0.009)
Criminal group controls						
Amigos dos Amigos (ADA)	$1.834^{\dagger}$	2.198^{*}	$-0.037^{\dagger}$	-0.031	-0.018	-0.028
	(0.971)	(0.971)	(0.020)	(0.020)	(0.030)	(0.030)
Comando Vermelho (CV)	0.048	0.032	0.000	-0.002	0.032	0.031
	(0.636)	(0.636)	(0.019)	(0.019)	(0.024)	(0.024)
Terceiro Comando Puro (TCP)	-1.083	-1.040	0.019	0.020	0.074^{*}	0.073^{*}
	(0.737)	(0.737)	(0.014)	(0.014)	(0.029)	(0.029)
Milícia groups	0.751	0.745	-0.012	-0.012	$-0.037^{\dagger}$	$-0.037^{\dagger}$
	(0.500)	(0.501)	(0.015)	(0.015)	(0.019)	(0.019)
Criminal conflict	0.184	0.298	-0.005	-0.005	0.015	0.010
	(1.406)	(1.398)	(0.023)	(0.023)	(0.029)	(0.028)
Observations	29,026	29,026	29,026	29,026	29,026	29,026
R^2	0.571	0.572	0.518	0.516	0.492	0.491
Independent variable	Binary	Count	Binary	Count	Binary	Count
Booth controls	✓	✓	✓	✓	✓	✓
Favela controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Polling station FE	✓	✓	✓	✓	✓	✓

Note: *Share favela* is logged. Standard errors are clustered at the polling station level: 1,338 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

a narrower candidate pool.

Columns 3 and 4 suggest that criminal corralling can influence vote choice for *milícia*-linked candidates. Where criminal corralling is present, there is a 6.6 percentage point increase in the probability that a candidate named in the *CPI das Milícias* wins the most votes at the voting booth, all else equal. This is also significant on the intensive margin; a one standard deviation increase in the residualized interaction term (SD = 1.43) corresponds to a 2.1 percentage point increase in the probability of winning the voting booth. Substantively, this is consistent with the *CPI das Milícias* findings. While the report documented over 171 *milícia*-dominated communities statewide with ties to public institutions, these ties were deeper in certain favelas than others, notably, the *Rio das Pedras* favela. In my data, *Rio das Pedras* consistently registers the among the highest levels of criminal corralling each year. This supports the theory that the presence of criminal corralling influences vote choice, and that higher levels of criminal corralling are even more effective.

Columns 5 and 6 provide further supporting evidence that criminal corralling can influence vote choice for both milícia- and trafficker-affiliated candidates alike. All else equal, there is a 6.8 percentage point increase in the probability that a candidate mentioned to *Disque Denúncia* will win the most votes in a voting booth when criminal corralling is present. Column 6 shows that higher levels of criminal corralling are also strongly correlated with the probability of victory for *all* alleged criminally-connected candidate. All results in Table 2 are robust to the exclusion of control variables (SI Table F1, p. 18).

When examining the point estimates for the criminal group controls, no clear patterns emerge. Columns 1-2 show that the gatekeeping effects are slightly blunted in favelas governed by ADA, and Columns 5-6 show that the TCP and milícias are positively and negatively correlated with vote choice, respectively. These point estimates may be puzzling, given that milícias are often associated with corralling behaviors (Cano and Duarte 2012). Several studies suggest that milícias have greater incentives to engage in elections than traffickers

Table 3: Criminal gatekeeping and corralling on voting, milícia-dominated favelas

	H1: Gatekeeping		**H2: Corralling**			
	N candidates		Vote choice (milícia)		Vote choice (traf.)	
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	-2.203^{**}	-0.345^{*}				
	(0.668)	(0.165)				
Criminal gatekeeping	-6.653^{**}	$-1.093^{\dagger}$				
	(2.389)	(0.590)				
Criminal corralling × Share favela			0.124^{**}	0.032^{**}	-0.115^{**}	-0.033^{*}
			(0.020)	(0.006)	(0.034)	(0.015)
Criminal corralling			0.384^{**}	0.100^{**}	-0.079	-0.071^{**}
			(0.052)	(0.012)	(0.091)	(0.019)
Share favela	-1.946^{**}	-1.949^{**}	0.015	0.016	-0.008	-0.008
	(0.563)	(0.567)	(0.012)	(0.012)	(0.006)	(0.006)
Observations	11,439	11,439	11,439	11,439	11,439	11,439
R^2	0.607	0.607	0.556	0.553	0.406	0.400
Independent variable	Binary	Count	Binary	Count	Binary	Count
Booth controls	✓	✓	✓	✓	✓	✓
Favela controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Polling station FE	✓	✓	✓	✓	✓	✓

Note: *Share favela* is logged. The outcomes in columns (3)-(4) and (5)-(6) are vote choice for milícia- and trafficking-affiliated candidates, respectively. Standard errors are clustered by polling station: 732 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

and have more resources to do so, given their connections to the state (Hidalgo and Lessing 2019; Paes Manso 2020).

In Tables 3 and 4, I condition the analysis by criminal group type (milícias and drug factions, respectively) to more closely examine the relationship between voting and gatekeeping or corralling within and across criminal groups. Columns 1 and 2 in both tables suggest that criminal gatekeeping is correlated with a restricted candidate pool in favelas governed by both types, although the effect size is larger and stronger for voters residing in trafficker-dominated favelas. Column 3-4 of Table 3 demonstrate that the presence of criminal corralling is highly correlated with a milícia-affiliated candidate winning in milícia-dominated areas (a 12.4 percentage point increase in the probability of winning the voting

booth), but Columns 3-4 of Table 4 show that corralling does not predict milícia-affiliated candidate victory in trafficker-dominated booths.

Next, in both tables, the dependent variable in Columns 5 and 6 captures vote choice for trafficker-affiliated candidates. Table 3 shows that there is a large and negative association between criminal corralling and vote choice for these candidates in milícia-dominated favelas, suggesting that these candidates are very unlikely to win votes in milícia-governed territories. The opposite is true within trafficker-dominated favelas; both the presence and the intensity of criminal corralling are significantly and positively associated with trafficker-affiliated candidate victory (Columns 5-6 of Table 4). A one standard deviation increase in the residualized interaction term (SD = 1.98) corresponds to a 8.1 percentage point increase in the probability that a candidate with alleged ties to traffickers wins in a trafficker-dominated voting booth.

These models serve the additional purpose of validating the criminal governance data, since we would expect candidates named in the *CPI das Milícias* to do well in milícia-dominated favelas, and for the "Vote share (traf.)" variable to be positively associated with corralling in trafficker-dominated areas. There are two important takeaways: even within criminal groups under the same banner, 1) there is substantial variation in gatekeeping and corralling, and 2) this variation maps onto the predicted outcomes, even within criminal groups. I find evidence supporting my theory that both milícias and traffickers can and do pursue both strategies to varying degrees.

Table 5 demonstrates that gatekeeping and corralling are also associated with lower electoral competition. I test the relationship between competition and gatekeeping, corralling, or both. All models suggest that the incidence and intensity of gatekeeping and corralling are significantly correlated with lower competition. Holding the logged *Share favela* at its mean (approximately one in twelve voters residing in a favela) and all else constant, the presence of criminal gatekeeping corresponds to a 37.5 percentage point increase in predicted

Table 4: Criminal gatekeeping and corralling on voting, trafficker-dominated favelas

	H1: Gatekeeping		**H2: Corralling**			
	N candidates		Vote choice (milícia)		Vote choice (traf.)	
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	-2.118^{**}	-0.467^{**}				
	(0.660)	(0.092)				
Criminal gatekeeping	-10.288^{**}	-2.485^{**}				
	(2.687)	(0.306)				
Criminal corralling × Share favela			0.002	0.000	0.107^{**}	0.041^{**}
			(0.006)	(0.001)	(0.032)	(0.005)
Criminal corralling			0.010	0.001	0.632^{**}	0.199^{**}
			(0.016)	(0.003)	(0.149)	(0.015)
Share favela	-2.028^{**}	-2.041^{**}	$-0.004^{\dagger}$	$-0.004^{\dagger}$	−0.001	−0.002
	(0.474)	(0.473)	(0.002)	(0.002)	(0.004)	(0.004)
Observations	13,449	13,449	13,449	13,449	13,449	13,449
R^2	0.587	0.592	0.746	0.746	0.563	0.584
Independent variable	Binary	Count	Binary	Count	Binary	Count
Booth controls	✓	✓	✓	✓	✓	✓
Favela controls	✓	✓	✓	✓	✓	✓
Criminal group controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Polling station FE	✓	✓	✓	✓	✓	✓

Note: *Share favela* is logged. The outcomes in columns (3)-(4) and (5)-(6) are vote choice for milícia- and trafficking-affiliated candidates, respectively. Standard errors are clustered by polling station: 904 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

vote concentration (Column 1), representing a significant tilting of the playing field. When repeating the same exercise for corralling (Column 3), the reduction in competition more than doubles, corresponding to a 59.4 percentage point increase in vote concentration. The least competitive local elections are those where both gatekeeping and corralling are present. Column 5 supports the claim that the mechanisms can be complementary, and their joint use has a compound effect on competition. However, both Columns 5 and 6 suggest that corralling has a stronger influence on electoral competition than gatekeeping.

These associations hold when considering electoral competition separately for milícia and trafficking groups. Table 6 shows that gatekeeping in milícia-dominated favelas is significantly correlated with lower levels of competition, corralling is even more so, and elections

Table 5: Criminal gatekeeping and corralling on competition

	H3: Electoral competition					
	HHI					
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	0.088**	0.018**			0.079**	0.003†
	(0.013)	(0.003)			(0.012)	(0.002)
Criminal gatekeeping	0.367**	0.078**			0.298**	0.006
	(0.053)	(0.012)			(0.052)	(0.007)
Criminal corralling × Share favela			0.114**	0.053**	0.084**	0.049**
			(0.017)	(0.010)	(0.018)	(0.011)
Criminal corralling			0.584**	0.241**	0.490**	0.233**
			(0.073)	(0.031)	(0.075)	(0.035)
Share favela	0.017*	0.017*	0.021**	0.020**	0.017*	0.020**
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Observations	29,026	29,026	29,026	29,026	29,026	29,026
R^2	0.681	0.683	0.685	0.697	0.688	0.697
Independent variable	Binary	Count	Binary	Count	Binary	Count
Booth controls	✓	✓	✓	✓	✓	✓
Favela controls	✓	✓	✓	✓	✓	✓
Criminal group controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Polling station FE	✓	✓	✓	✓	✓	✓

Note: HHI and *Share favela* are logged. Standard errors are clustered at the polling station level: 1,338 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

are least competitive when both are present. Table 7 suggests that both gatekeeping and corralling are associated with lower competition in trafficker-dominated favelas, but to a lesser extent than for milícias, especially for corralling. In trafficker-dominated favelas, corralling is subsumed by the stronger effect of gatekeeping on competition when both are present (Colums 5-6 of Table 7), whereas the inverse is true for milícia groups. This is not to say that traffickers cannot reduce competition or sway vote choice, but rather, they may not be able to reduce competition to the extent that milícia groups are, which is in line with existing evidence about milícia's connections to the state's apparatus. Though out of the scope of this paper, future work could probe the results in Tables 3-4 and Tables 6-7 to explore whether the level of gatekeeping or corralling is a function of criminal group resources and

Table 6: Criminal gatekeeping and corralling on competition, milícia-dominated favelas

	H3: Electoral competition					
	HHI					
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	0.108**	0.020**			0.088**	0.004*
	(0.016)	(0.006)			(0.018)	(0.002)
Criminal gatekeeping	0.381**	0.070**			0.299**	0.007
	(0.067)	(0.024)			(0.074)	(0.009)
Criminal corralling × Share favela			0.169**	0.088**	0.152**	0.085**
			(0.033)	(0.013)	(0.037)	(0.014)
Criminal corralling			0.606**	0.374**	0.539**	0.369**
			(0.081)	(0.019)	(0.089)	(0.019)
Share favela	0.021*	0.020*	0.022*	0.020*	0.019*	0.019*
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
Observations	11,439	11,439	11,439	11,439	11,439	11,439
R^2	0.721	0.720	0.726	0.746	0.729	0.747
Independent variable	Binary	Count	Binary	Count	Binary	Count
Booth controls	✓	✓	✓	✓	✓	✓
Favela controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Polling station FE	✓	✓	✓	✓	✓	✓

Note: *HHI* and *Share favela* are logged. Standard errors are clustered by polling station: 732 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

opportunity costs.

As with any natural experiment, there is a risk that pre-treatment confounders may drive the results. The stability of the results after controlling for favela- and voting-booth-level factors suggests that existing explanations related to poverty, demographics, or specific criminal groups – even between favelas – do not fully account for the differences in voting amid gatekeeping and corralling. Further, the inclusion of yearly and polling-station-level fixed effects minimizes concerns that key differences between polling stations or years are driving the relationship between voting behavior and gatekeeping or corralling.

Table 7: Criminal gatekeeping and corralling on competition, trafficker-dominated favelas

	H3: Electoral competition					
	HHI					
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	0.089**	0.015**			0.093**	0.012**
	(0.019)	(0.003)			(0.019)	(0.003)
Criminal gatekeeping	0.425**	0.081**			0.385**	0.044**
	(0.084)	(0.009)			(0.072)	(0.009)
Criminal corralling × Share favela			0.104**	0.025**	0.061*	0.007
			(0.030)	(0.005)	(0.027)	(0.007)
Criminal corralling			0.594**	0.153**	0.446**	0.089**
			(0.126)	(0.016)	(0.108)	(0.020)
Share favela	0.026*	0.028*	0.033**	0.033**	0.026*	0.030*
	(0.011)	(0.012)	(0.012)	(0.012)	(0.011)	(0.012)
Observations	13,449	13,449	13,449	13,449	13,449	13,449
R^2	0.717	0.726	0.719	0.731	0.724	0.732
Independent variable	Binary	Count	Binary	Count	Binary	Count
Booth controls	✓	✓	✓	✓	✓	✓
Favela controls	✓	✓	✓	✓	✓	✓
Criminal group controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Polling station FE	✓	✓	✓	✓	✓	✓

Note: *HHI* and *Share favela* are logged. Standard errors are clustered by polling station: 904 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

Robustness and validity checks

I conduct a range of additional tests to probe these results, beginning with sensitivity tests of the independent variables. The validity of the independent variables depends on accurate and unbiased reporting to *Disque Denúncia*. I assume that gatekeeping and corralling are underreported, as is the case with most reported crime or malfeasance. This means that my measures are likely conservative estimates of their true incidence, and may be undercounting gatekeeping or corralling in communities where voters do not call. A conservative interpretation of the main independent variables is that they are identifying the favelas where

gatekeeping or corralling is most egregious.[27] If true, a valid interpretation of the main findings is about incidence of a high enough level of gatekeeping or corralling that prompts voters to call. The following tests take these concerns into account.

One threat to inference is that the independent variables are identifying favelas which are simply more engaged in politics, and this is a story about favela politics rather than criminal politics. I replace the corralling and gatekeeping measures with measures of general electoral activity in favelas.[28] General electoral activity is inversely correlated with the number of candidates receiving votes and positively correlated with competition, consistent with the main findings (SI Table F2, p. 19). This suggests that favelas that are more active in politics might also have a tendency to vote in bloc. However, these point estimates are much smaller than those in the main results, suggesting that electoral activity is not singularly responsible for the main findings. Further, there is an inconsistent relationship between favela electoral activity and vote choice for criminally-affiliated candidates, suggesting that general electoral activity does not explain all the variation in voting outcomes.

I then consider whether the electoral outcomes result from gatekeeping and corralling in general, rather than their specific use by criminal groups (SI Table F3, p. 19).[29] These results are similar in significance and direction to the main findings, but much smaller in magnitude across all outcomes. I interpret these findings as suggesting that gatekeeping and corralling, when practiced by other brokers, can be successful – but not to the same extent as when practiced by criminal groups. This supports my argument that territorial control gives criminal groups efficiency advantages over competing brokers.

Another concern is that the word lists used to generate the variables for criminal gate-

27. Public security professionals suggest that the population calls *Disque Denúncia* to complain only when "things are really bad." (Interview 6).

28. A complaint about general electoral activity states, "Candidates *NAME* and *NAME* are holding parades by car, with boom boxes playing very loud music" (Complaint 4).

29. A complaint about general corralling states: "Near *PRIMARY SCHOOL*, there are close to fifteen people buying votes. They are paying R$70 to voters." (Complaint 5).

keeping or criminal corralling are arbitrary, incomplete, or biased.[30] I cross-validate my primary independent variables with others less reliant on researcher choice. I fit a structural topic model on the corpus of *Disque Denúncia* calls (Roberts, Stewart, and Tingley 2019), detailed in SI Section A2 (pp. 2-6). I measure whether "Topic: Elections" calls occur for each favela-year observation, then re-estimate the models (SI Table F4, p. 20). The results for H1 and H3 are attenuated but statistically significant and consistent with the main results, but the results for H2 are inconsistent and dependent on how the STM variable is constructed.

One might also be concerned that favela residency is not an adequate proxy for criminal governance. Using the *Disque Denúncia* corpus, I address this by filtering calls for those that mention a common form of criminal governance: service provision, especially "protection" or security provision (See SI Section C.2, pp. 11-12). I restrict the sample of favelas to those where at least one call mentioned both a criminal group *and* service provision and re-estimate Equation 1; the magnitude and significance parallel those in the main results (SI Table F5, p. 21).

I then conduct sensitivity tests of the dependent variables. I operationalize gatekeeping as the number of candidates receiving five or more votes, and find support for my hypotheses (SI Table F6, p. 22). For corralling, one concern is that the *CPI das Milícias* vote choice outcome omits many candidates whose links to milícias emerged after 2008. I restrict the sample to the post-*CPI* period and re-estimate the models, using the *CPI das Milícias* outcome (Columns 3-4). I then filter my predictor variable to milícia-affiliated candidates named in *Disque Denúncia* but *not* named in the *CPI*, and re-estimate models for 2012-2020 (Columns 5-6). Though attenuated, given the removal of many candidates and one election year, there is a positive and significant association between criminal corralling and

30. For example, it is common to use euphemisms instead of the faction names that frustrate attempts to classify text.

vote choice for milícia-linked candidates. Then, as a substitute for HHI, I measure the share of votes the leading candidate in the booth receives (Columns 7-8). Both gatekeeping and corralling are significant predictors of winning by a larger margin.

The main argument focuses on favelas with consolidated territorial control; an important scope condition is that criminal groups can only effectively gatekeep and corral in such areas. I restrict the sample to favelas that experienced contested authority prior to the election in SI Table F7 (p. 23).[31] The coefficients for gatekeeping have a large and negative association with the number of candidates receiving votes, which at first blush may appear at odds with my argument. However, instead of the gatekeeping relationship described in the theory section, I contend that there are large, unobserved decreases in the supply of candidates attempting to campaign in these areas, as few want to enter an active war zone.[32] There are null results for the models focused on corralling and inconsistent results for electoral competition, which further supports the claim that voting patterns under contested governance look different than the dynamics described in this paper.

Lastly, my design increases the precision in measures of favela voting, which was previously done by classifying polling stations as inside or outside of a favela's buffer zone. My approach reveals that favela and asphalt residents intermingle at the voting booth, even in polling stations closest to favela boundaries. I restrict the sample to voting booths "inside" the 500 meter buffer of the closest favela and find statistically significant support for all of my hypotheses (SI Table F8, p. 24).

These results show that criminal gatekeeping is strongly associated with reduced contestation, criminal corralling is strongly associated with vote choice for both milícia and trafficker-affiliated candidates, and both mechanisms are strongly associated with lower competition. These findings hold when conditioning on criminal group type and governance proxies, and

31. See SI Section C1, pp. 8-11 for how I identified those favelas.

32. These patterns are consistent with the theories and evidence in Gallego (2018b) and Gutiérrez Sanín (2015).

testing the sensitivity of dependent and independent variables. While this analysis cannot, with certainty, separate the effects of favela residency from living under criminal governance, these tests show that within-favela variation and the clientelistic voter mobilization strategies associated with favelas (SI Tables F2 and F3 on p. 19) alone do not explain the persistent relationship between voting and criminal gatekeeping or corralling.

Discussion

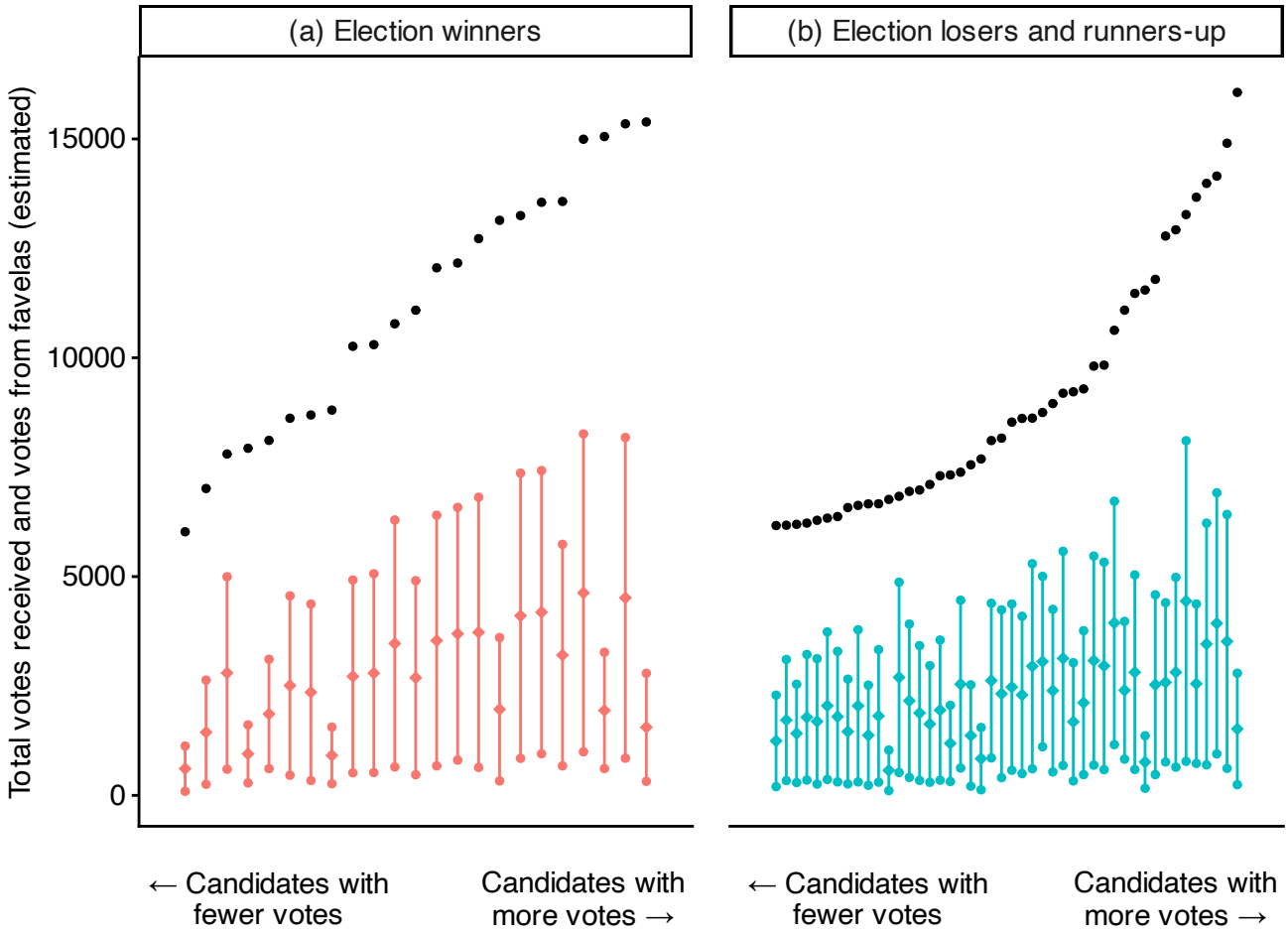
Overall, these results establish a strong inverse relationship between criminal gatekeeping and corralling and electoral competitiveness at the voting-booth-level. What is the impact of gatekeeping or corralling on elections beyond the voting booth? It is challenging to calculate how many votes candidates get from favelas, due to the voter assignment procedure. Even candidates are uninformed; a campaign manager admitted that the geographic fuzziness in voter assignment made it too difficult for their team to observe campaign performance below the neighborhood level (Interview 7).

Despite these data limitations, I estimate the favela votes that anonymized *marginal* candidates received, using the 2016 election as a test case. Figure 2 plots the total votes received for 69 marginal candidates (23 winning and 46 losing candidates) alongside an estimated range of favela votes received.[33] I define a candidate as *marginal* if they received no more votes than the highest-earning losing candidate (16,064 votes), but as many or more than the lowest-earning winner (6,023 votes). Given their precarious position, I expect that marginal candidates are most likely to benefit from gatekeeping and corralling. My estimates show that marginal candidates receive, on average, 24% of their votes from criminally governed favelas.

I estimate back-of-the-envelope calculations of whether the favela votes were decisive in

33. These estimations are described in SI Section E, pp. 15-16.

Figure 2: Favela votes as a share of candidates' total votes, 2016



Note: Marginal candidates are anonymized and sorted by votes received. The points at the top represent the total votes earned, and those between the errorbars represent the estimated upper, lower, and mean favela votes received.

securing victory for the marginal winners. The open list system complicates the construction of counterfactual election outcomes, but I draw some aggregate conclusions. First, six winners (12% of the city council) definitely would not have won without favela votes, and the redistribution of these votes to other candidates would have led them to lose. Second, six others won by even narrower margins due to 2,000 or fewer favela votes. Even a partial redistribution of their favela votes – or a change in the total votes received by their party – would have cost them the race.

In sum, 12 winners (24% of the city council) received approximately a quarter or more of their votes from favelas, and could have lost without the support of all or some favela voters, often at margins as thin as 2,000 votes. These 12 candidates who eked out a victory were also more likely to be named in *Disque Denúncia* complaints than other winners. In one egregious case, voters complained about a candidate 53 times; nearly all complaints mentioned the use of criminal brokers.

Conclusion

Politicians all over the world call on criminal groups to help win elections. Criminal groups that govern can effectively marshal their territorial control to influence voters, which candidates seek. I argue that politicians hire criminal groups as brokers, paying for their capacity to deliver votes by gatekeeping rival candidates and corralling voters. I offer empirical evidence of both gatekeeping and corralling, showing a persistent and strong relationship between gatekeeping and reduced contestation, and corralling and vote choice, for both milícia- and trafficker-affiliated candidates. This paper also demonstrates that both mechanisms are associated with reduced electoral competition, and explores the use and efficacy of either mechanism by different criminal group types.

This paper reveals how criminal groups derive power from border and internal territorial control, and use it to gatekeep and corral. Past research argues that criminal groups that want to influence electoral outcomes strategically use bribes or vote buying (Albarracín 2018; Arias 2006, 2017; Blume 2022; Gay 1999; Gutiérrez Sanín 2015) or violence (Acemoglu et al. 2013; Alesina et al. 2019; Córdova 2019; Daniele and Dipoppa 2017; Dell 2015; Steele 2011; Trejo and Ley 2021). But this is only part of the story. Undergirding their ability to deploy bribes or violence effectively is their control over territory. Criminal groups leverage border control to gatekeep, limiting candidates' access to voters to those that pay up. Gatekeeping

effectively prunes rival candidates from the candidate pool. Criminal groups corral by issuing rewards and threats to voters that are rooted in their internal control over the neighborhood's residents. These rewards and threats influence vote choice on Election Day. Future research should investigate the medium- and long-term consequences of these electioneering tactics, and which candidates use them.

Rio de Janeiro is a microcosm in which we can observe criminal governance and candidate competition at the sub-municipal level. While this paper focuses on local dynamics, it also generates predictions about criminal governance and electoral politics in other contexts. I highlight the importance of two governing behaviors, by way of territorial control, that explain why we observe criminal groups succeeding as brokers in Rio de Janeiro and beyond. Future research should also investigate how criminal brokers deliver votes – if at all – in different contexts where criminal groups have little to no territorial control or presence.

An estimated 77 million people experience criminal rule in Latin America alone (Uribe, Lessing, Schouela, and Stecher 2025). My results shed light on the implications of criminal governance for electoral politics and fighting crime in such contexts. While the dominant approach to fighting crime overwhelmingly focuses on policing, there are reasons to be skeptical that policing alone is sufficient, given that preferences for law enforcement policies are responsive to electoral politics (Flom and Post 2016; Holland 2013; Laterzo 2024; Novaes 2024), and the many symbiotic relationships between state actors and organized crime (Barnes 2017; Flom 2022; Trejo and Ley 2020). How can the government be expected to fight crime if individual policymakers – often those who write the rules – benefit electorally from criminal connections?

This paper provides a theory that helps understand politicians' electoral motives for partnering with criminal groups. In doing so, it encourages skepticism about crime-fighting strategies that only focus on law enforcement instead of the broader governance challenges associated with criminal rule. These conclusions inform our understanding of elections and

democracy in contexts where criminal groups govern. When criminal groups wield such control over voters, it is not surprising that politicians will be eager to capitalize on this influence at the ballot box.

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How criminal governance undermines elections
Supplemental Information

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A Description of independent variables

A.1 Main variables: criminal gatekeeping and criminal corralling

I calculate count and binary variables for 1) organized crime, 2) general electoral activity, 3) gatekeeping, and 4) corralling, at the favela-election-year level for the 2008, 2012, 2016, and 2020 Rio de Janeiro municipal elections. Using the 2016 favela shapefile, published by the Rio de Janeiro municipality and managed by the municipal organization *Instituto Pereira Passos*, I classify *denúncias* as inside a favela if they fall within a narrow 100 meter buffer of its perimeter. Then, using text data from 42,889 *Disque Denúncia* calls that occurred inside favelas during the campaign,[1] I measure if each individual complaint call (*denúncia*) contains words that pertain to each topic, shown in Table A1. Then, I construct three additional *denúncia*-level indicators: 1) general criminal electoral activity, 2) criminal gatekeeping, and 3) criminal corralling, which take on a value of 1 only if organized crime keywords are mentioned in the same complaint as election, gatekeeping, or corralling keywords, respectively.

To construct the variables used for the main analyses and reported in Table 1, I aggregate the *denúncia* topics by favela-election-year and generate 1) binary indicators of their presence and 2) count variables of their extent. Figures A1 and A2 print a random sample of the translated *denúncias* that contained both organized crime words *and* gatekeeping or corralling words, respectively.

A.2 Alternative independent variables: structural topic model

To cross-validate the measures of criminal gatekeeping and corralling that rely on the key words shown in Table A1, I fit a structural topic model on the campaign season *Disque Denúncia* calls (Roberts et al. 2019). I use Mimno and Lee's (2014) algorithmic approach to specify the number of topics, rather than relying on an arbitrary researcher-defined number. One topic (hereafter, "Topic: Elections") was clearly about elections, as shown by the most common words in Figure A3. Approximately 1.75% of the corpus of *denúncias* belongs to this topic during this time period, which makes sense, given that people usually call *Disque Denúncia* to complain about disturbances, low-level crime, or events unrelated to politics.

I then create a few variables to measure the prevalence of "Topic: Elections" in favelas for each year. Using the `stm` R package and the notation shown in Roberts et al. (2019), I calculate for each *denúncia* i and topic j the $\theta_{i,j}$ coefficient (the proportion of "Topic j: Elections") in *denúncia* i). I aggregate the $\theta_{i,j}$ data across *denúncias* to calculate three variables at the favela-election-year level, used as alternate independent variables and reported

1. I define the campaign period as beginning on July 1 and ending one week after the first-round election date, for approximately 90 days. This includes the formal 45-day campaign period, 45 days prior (when campaigning is not legal but is widespread) and the days after the election.

Figure A1: Criminal gatekeeping *denúncias*

Original Text	Translation
Em todo morro citado, constantemente, desde inicio das campanhas eleitorais, traficantes (nao identificados) sao vistos retirando placas entre outras formas de propagandas eleitorais de outros candidatos que nao sejam os de sua preferencia ordem foi passada por [redacted] (nao caracterizada), irma os traficantes [redacted], que ja encontram-se preso qual apoia candidato vereador conhecido como [redacted] permite que apenas ele divulgue sua candidatura na comunidade.	Throughout the mentioned favela, since the start of the electoral campaigns, drug traffickers (unidentified) have been seen removing signs and other forms of campaign propaganda from candidates who are not their preferred choices. The order was given by [redacted] (not further described), sister of the trafficker [redacted] who is already in prison and supports the city council candidate known as [redacted] Only [redacted] is allowed to promote his candidacy in the community.
No endereco citado, na associacao [redacted], pode ser visto presidente [redacted] que era nomeada da prefeitura cedeu um terreno no qual funcionava [redacted] para [redacted] esposa do traficante vulgo [redacted], para fazer sua casa. [redacted] deu um quiosque na [redacted], uma barraca de praia. Na associacao trabalha [redacted] filha do traficante [redacted] (nao identificadas). Ha sobrinhos irmao de [redacted] que chefe da obra na comunidade. [redacted] sobrinha de [redacted], proibiram candidatos de realizam campanha no morro [redacted].	At the mentioned location, in the residents' association of [redacted], the president, [redacted] who was appointed by the city government, handed over a plot of land—previously used as the [redacted] center—to [redacted] wife of the drug trafficker known as [redacted] to build her house. [redacted] also gave her a kiosk at [redacted]. Within the association, [redacted] daughter of the trafficker [redacted] and her unidentified sisters work. There are also nephews and the brother of [redacted] who is the construction chief in the community. [redacted] niece of [redacted] has prohibited any candidates from campaigning in [redacted].
No endereco mencionado, proximo estacao de trem de [redacted], acesso pela rua [redacted], localiza-se uma casa de dois andares, onde reside [redacted], vulgo [redacted], um dos assessores do candidato [redacted] tem envolvimento com os traficantes [redacted], este ultimo oriundo da [redacted] juntos decidiram proibir entrada de outros candidatos em [redacted], permitindo apenas que [redacted] faca campanha eleitoral. Os moradores da regiao estao sendo coagidos votarem nesse candidato. Em troca, [redacted] pagou um valor de 30.000,00 ao trafico de drogas, para fazer sua campanha.	At the mentioned address, near [redacted] train station, with access via [redacted] Street, there is a two-story house where [redacted] known as [redacted] lives. He is an advisor of city council candidate [redacted] is involved with drug traffickers [redacted] and [redacted] the latter being originally from [redacted]. Together, they decided to prohibit the entry of any other candidates into [redacted], allowing only [redacted] to campaign there. The region's residents are being coerced into voting for this candidate. In exchange, [redacted] paid R$30,000 to the drug trade to fund his campaign.
Na rua citada, esquina com a rua [redacted], no morro [redacted], na comunidade [redacted], situa-se um barraco, utilizado como esconderijo do traficante [redacted], onde será realizada uma reunião na quinta-feira [redacted], por volta das 20h, com chefes do tráfico de várias comunidades e os líderes comunitários da [redacted] e [redacted], junto a vários candidatos a vereador na região [redacted]. Eles estão exigindo que os candidatos paguem taxas no valor de r$ 10.000,00 (dez mil reais), para que possam fazer campanhas na [redacted] e nas favelas vizinhas. Nesta quarta-feira [redacted], traficantes, armados, proibiram qualquer tipo de campanha na localidade mencionada. Obs.: citados, não descritos ou não identificados.	On the aforementioned street, at the corner with [redacted] Street, on [redacted], in the [redacted] community, there is a shack used as a hideout by the trafficker [redacted] where a meeting will be held on Thursday [redacted] around 8:00 p.m. with drug-trafficking chiefs from several communities and the community leaders of [redacted], [redacted] and [redacted] together with several candidates for city council in [redacted]. They are demanding that the candidates pay fees of R$10,000.00 (ten thousand reais) in order to campaign in [redacted] and the neighboring favelas. On Wednesday [redacted], armed traffickers prohibited any kind of campaigning in the aforementioned area. Note: persons named but not otherwise described or identified.

Figure A2: Criminal corralling *denúncias*

Original Text	Translation
Informa que, na favela citada, em frente curva do ha um ciep (centro integrado de ensino publico), onde hoje [redacted] partir das 00h havera um baile funk, patrocinado pelo vereador [redacted], que foi eleito, nas eleicoes do ano 2008, para agradecer apoio dos traficantes em sua candidatura. Afirma que ele foi eleito por meio de compra de votos com apoio do trafico de drogas. Traficantes estarao no local. Pede averiguacao.	In the mentioned favela, in front of the curve where there is a *CIEP* (*Centro Integrado de Ensino Público* - Integrated Public Education Center), today [redacted] starting at midnight, a *baile funk* (funk party) will be held, sponsored by city councilman [redacted] who was elected in the 2008 elections. The event is meant to thank drug traffickers for supporting his candidacy. The source claims that he was elected through vote-buying with the support of drug traffickers. Traffickers will be present at the event. *Requests investigation.*
Na rua citada na rua [redacted], acessos comunidade da [redacted], candidato vereador do partido [redacted], [redacted] esta utilizando funcionarios (nao identificados) da administracao regional, para construirem diversos quebra-molas nestas ruas pedido dos trafico local com intuito de obter votos dos moradores locais citado, era quem comandava regiao administrativa.	On the mentioned street, [redacted], which provides access to the [redacted] community, a city council candidate from the [redacted] party, [redacted], is using employees (unidentified) from the regional administration to build multiple speed bumps on these streets at the request of local drug traffickers. The aim is to gain votes from local residents. The source states that [redacted] as the one who previously controlled the regional administration.
No endereco citado, proximo [redacted], situa-se bar (nao denominado), onde encontram-se dois individuos (nao identificados), armados, levando crer serem milicianos, distribuindo panfletos aos eleitores.	At the mentioned address, near [redacted], there is a bar (unnamed) where two individuals (unidentified), armed and believed to be militia members, are distributing campaign pamphlets to voters.
Na rua citada, proximo ao [redacted] bpm, localiza-se um sacolao, onde podem ser vistos, aos finais de semana partir de 13h, os milicianos [redacted] de vulgo [redacted] policial militar [redacted] que fica apaisano com um fuzil dentro de uma capa de violao, ameacando os moradores pedindo voto para [redacted] de nº [redacted].	On the mentioned street, near the [redacted] (*Military Police Battalion*), there is a produce market, where, on weekends from 1 p.m. onwards, militia members [redacted], known as [redacted] and [redacted] a military police officer, can be seen. Officer [redacted] is in plain clothes, carrying a rifle inside a guitar case, intimidating residents and demanding votes for candidate [redacted] campaign number [redacted]

Table A1: Topic keywords

Topic	Keywords	Keywords (portuguese)
Organized crime	*Amigos dos Amigos*,[1] Command, *Comando Vermelho*, *CV*, Drugs, Faction, War, *Milicia*, Rival, Traffic-, *TCP*, *Terceiro Comando*	AMIGOS DOS AMIGOS, COMANDO, COMANDO VERMELHO, CRIME ORGANIZADO, CV, DROGA, FACCAO, GUERRA, MILICIA, RIVAL, TRAFIC, TCP, TERCEIRO COMANDO
Election (general)	(To) Canvass, City council, Campaign, Candidat-, Elect-, Voter, Pamphlet, Politic-, Advertisement, Ballot box, City councillor, Vote	BOCA DE URNA, CAMPANHA, CANDIDAT, ELEIC, ELEITOR, PANFLETO, POLITIC, PROPAGANDA, URNA, VEREADOR, VOTO
Gatekeeping[2]	Access, Authoriz-, Enter, Expel-, Permiss-, Permit-, Prohib-	ACESSO, AUTORIZA, ENTRA, EXPULS, PERMISS, PERMIT, PROIB,
Corralling	(To) Canvass, Vote buying, Pamphlet, Advertisement, Ballot box, Vote	BOCA DE URNA, COMPRA DE VOTO, PANFLETO, PROPAGANDA, URNA, VOTO

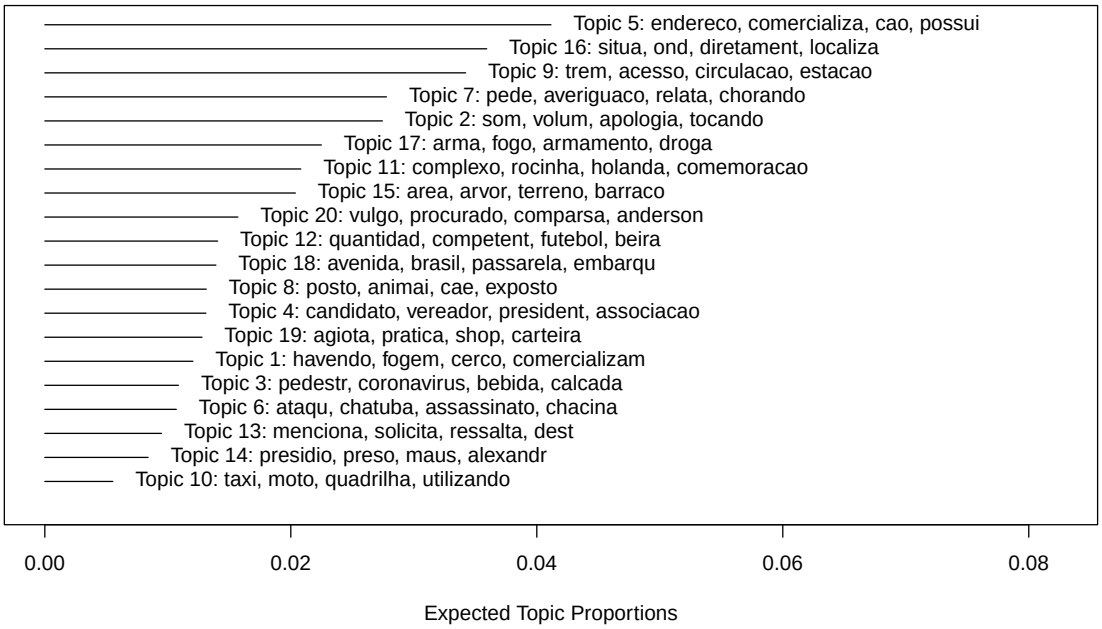
[1] Italicized names on this list are directly translated, as they are the full names or abbreviations of criminal organizations.

[2] The *denúncia*-level *Gatekeeping* indicator is constructed by measuring the joint presence of the gatekeeping keywords *and* the "Election (general)" keywords. For example, if a *denúncia* contains the words "candidate" and "authorize", it is coded as a "1" if it just contains "authorize" but no other election-related words, it is coded as a "0."

in Table F4:

- **STM binary:** An indicator variable if any *denúncias* per favela-year have "Topic: Elections" as their most prevalent topic. Mathematically, if any $\theta_{i,Elections} = \max_j \theta_{i,j}$
- **STM count:** A count variable of the number of *denúncias* per favela-year that have "Topic: "Elections" as their most prevalent topic. This is expressed as $\sum_i \mathbb{1}(\theta_{i,Elections} = \max_j \theta_{i,j})$ for each favela-year.
- **STM $\geq$ 10:** A count variable of the number of *denúncias* per favela-year that are at least a 10% match to "Topic: Elections." This is expressed as: $\sum_i \mathbb{1}(\theta_{i,Elections} \geq 0.1)$ for each favela-year.

Figure A3: STM top topics



Note: "Topic 4" is the Elections topic. From left to right, the highest-ranking words in this topic are *candidate, city councillor, president*, and *association*. The Mimno and Lee (2014) algorithm fit a total of 55 topics to the *Disque Denúncia* corpus. This figure only shows the 20 most prevalent topics for readability.

B Description of outcome variables

B.1 Description of vote choice outcomes

I construct three variables to measure the vote choice outcomes. For all of them, I begin by identifying the candidate that won the most votes for each ballot-box-year, using the data from the Supreme Electoral Court (TSE). Then, I generate a binary variable ("Vote choice") that takes on a value of 1 if that particular candidate appeared in any of the three lists of alleged criminally-affiliated candidates.

- **Vote choice (milícia)**: The first variable uses the *CPI das Milícias* document as the basis for the list of criminally-affiliated candidates. I collect all city council candidate names mentioned in the document, including those who ran in future election-years but did *not* run in 2008 (the investigation occurred in 2008). The names, and the election-years, are printed in Table B1. This extends the work done by Hidalgo and Lessing (2019), who focus on candidates named in the report who were successfully elected to the assembly. Not included in this list are pre-candidates for city council

who dropped out of the race before election day, such as "Jerônimo Guimarães Filho."

- **Vote choice (all)**: The second variable uses the *Disque Denúncia* data as the basis for the list of criminally-affiliated candidates. I filter all *denúncias* for the stem words "CANDIDAT-," and "VEREADOR-" (candidate and city councillor, respectively), extract, and hand-code the names of all councillors to generate this list. Once I standardized the written names with the full legal names from the TSE data, I applied a coding rule that candidates had to be mentioned in *Disque Denúncia* two or more times in order to be included in the final candidate list. A qualitative reading of these complaints suggested that candidates only mentioned once were complained about for more benign reasons (loud music, annoying advertisements) compared to the candidates complained about systematically, which were more likely to be for reasons related to organized crime.

- **Vote choice (traf.):** The third variable filters the list of candidates used to generate the **Vote choice (all)** variable to *only* those candidates named for affiliation with trafficking groups.

Table B1: City council candidates mentioned in the *CPI das Milícias*

Name	**Election years**
CARMEN GLORIA GUINANCIO GUIMARÃES (TEIXEIRA)*	2008, 2012
JOSINALDO FRANCISCO DA CRUZ	2008
ELTON JORGE HAUAT	2008, 2012, 2016
LUIZ ANDRE FERREIRA DA SILVA	2008
CRISTIANO GIRÃO MATIAS	2008
LUIZ MONTEIRO DA SILVA	2008
JOSÉ FERNANDO MORAES ALVES	2008, 2012
ALEXANDRE BATISTA CERRUTI	2008
SILVIA REGINA PINTO PONTES	2008
MARCELO HENRIQUES BAPTISTA	2008, 2012, 2016, 2020
FRANCISCO FELIX VALENTE	2008
WLADIMIR GONÇALVES MAMEDE	2008
JAIR BARBOSA TAVARES	2008, 2016, 2020
JURANI FERREIRA DA ROSA	2008
MARCIO AMARAL DE CASTILHO	2008
JORGE MAURO DOS SANTOS SILVA	2008, 2012, 2016, 2020
MARCELO ZATURANSKY NOGUEIRA ITAGIBA	2012
JOÃO FRANCISCO INÁCIO BRAZÃO	2008, 2012, 2016

*This candidate used different legal names depending on the election year.

For all three candidate name lists (*milícia*, *all*, and *traf.*), I generate the outcome variables reported in Table 1 by using the full legal name as a crosswalk to link the name lists to the

TSE electoral returns. If a candidate on any of these three lists was the most voted candidate for a ballot-box-year, the **Vote choice ( )** variable takes on a value of 1 for that observation, if not, 0.

B.2 Description of vote concentration outcome

For the outcome measuring electoral competition, I use the Herfindahl-Hirschman Index (HHI), a commonly used measure of vote concentration in electoral studies. For every voting booth b, I construct the HHI by summing the share of votes received (v_c) of all candidates $c \in \{1, ...n\}$ who ran for city council in a given year:

$$HHI_b = \sum_{c=1}^{n} v_c^2$$

The support of HHI_b is between 0 and 1, with larger values corresponding to a larger share of the votes going towards one candidate.

C Description of control variables

All favela-level variables, including criminal governance, control for the most-represented favela at the voting booth. If there are residents from multiple favelas that vote at the same booth (and there often are), the favela controlled for is the one with the highest share of voters at the booth.

C.1 Criminal group controls

As with the main independent variables, I use the corpus of *denúncias* to build a panel database at the favela-month level from 2008-2020. I create four indicators and assign each *denúncia* a value of 1 if it contains the words for the topics shown in Table C1. I then aggregate *denúncias* at the favela-month level and code a favela's monthly criminal governance using the following process:

1. If only one faction is identified as present in a favela for the entire time series (2008 - 2020), I code the favela as governed by that faction for the months when a faction is named. These favelas comprise 41% (445) of all 1,075 favelas.

2. For all other favelas, I hand-code the monthly governing faction, generating individual favela timeseries files with *denúncia* texts to efficiently hand-code large quantities of information. When multiple faction names are present in the same favela per month, I resolve this in one of three ways:

   (a) **Only one faction**: There is only one governing faction, despite multiple being mentioned. For example, a rival faction may be mentioned in a *denúncia* if the

Table C1: Criminal governance topic keywords

Group	Keywords	Keywords (pt)
CV	C.V., CV Faction, Red Command, CV Criminal	C.V., FACÇÃO CV, COMANDO VERMELHO, CRIMINOSA CV, CRIMINOSO CV
ADA	A.D.A., ADA Faction, Friends of Friends,* ADA Criminal	A.D.A., FACÇÃO ADA, AMIGOS DOS AMIGOS, AMIGO DOS AMIGOS, CRIMINOSA ADA, CRIMINOSO ADA
TCP	TCP,* Third Pure Command,* TCP Criminal	TCP, T.C.P., TERCEIRO COMANDO PURO, TERCEIRO COMANDO, CRIMINOSA TCP, CRIMINOSO TCP
Milícia	Milícia,* Milicia-	MILICIA, MILÍCIA

*Indicates that alternate spellings of the same term and common spelling errors are included as keywords. Diacritics are removed so that both spellings with and without accent marks will be tagged to a faction.

local governing faction is preparing to threaten them or engage in another way. When this is the case, I edit the favela-month so only the governing faction's indicator shows their presence.

(b) **Multiple factions**: Though uncommon, factions have been documented cooperating and co-governing certain communities. One of the most common alliances is the TCP faction and *milícias*, especially in later years of the time series. For these favela-months, I edit the code so that both governing factions' indicators show their presence.

(c) **Criminal conflict:** I code favela-months as not pertaining to any faction and in a "Criminal conflict" state when the *denúncias* report large-scale conflict between at least two factions in the favela. The *denúncias* often make explicit a faction's goal to take over or retake a favela, and demonstrate widespread violence and terror.

3. Then, I impute the monthly favela-level data over time. Though *Disque Denúncia* complaints are common, monthly-level data is high-frequency enough that often months go by in a favela where there are no complaints about a criminal group, but this does not mean the criminal group has left. I impute the criminally governing faction up to 24 months (2 years) into the future and 12 months in the past. When there are infrequent complaints about a sole faction in a favela, I impute up to five years of missingness in between observations.

4. I then cross-validate my data in two ways. First, I recruited three graduate student research assistants that are familiar with the context and are native Portuguese speakers. I gave them a random sample of favelas and asked them to hand-code the entire

time series, resolving all multiple mentions in one of the three above ways. Their favela-month code was a match to mine 92% (lower bound) to 100% of the time.

Second, I compare my code to the 2019 cross-section from *Fogo Cruzado*, available here: https://erickgn.github.io/mapafc/. *Fogo Cruzado*, a civic-tech organization, also uses the *Disque Denúncia* data to map criminal governance in Rio de Janeiro, but unlike me, approaches the problem of territorial boundaries inductively rather than deductively. *Fogo Cruzado* defines criminally governed territory depending on the concentration of *denúncias* rather than the administrative favela boundaries, meaning that there are a different number of criminally-governed polygons in their dataset than in mine. On average, when comparing my data from 2015 - 2020 to their 2019 annual cross-section, my data is a 68% match, but represents a substantial improvement over the *Fogo Cruzado* data, given that my data was hand-coded and validated by multiple readers, and is more granular (at the monthly rather than yearly level).

The implications of these coding differences are that there are far more areas coded as *"Em disputa"* ("Criminal conflict") by *Fogo Cruzado* than by me. Unfortunately, the details of their methods are not available online or via email,[2] but it appears that when multiple factions were mentioned inside a favela-year, they classify it as *"Em disputa."* I resolved some of these cases in my dataset by reading the complaints and hand-coding (See Step 2(a)-(c) above), which allowed me to eliminate false positives for the "Criminal conflict" category.

As an example, one favela (let's call it *Favela A*) is coded as *"Em disputa"* in *Fogo Cruzado*'s 2019 map, yet is coded as under the control of ADA (*Amigos dos Amigos*) drug trafficking faction for all months in 2019 in mine. *Disque Denúncia* calls reveal that *Favela A* was involved in a turf war in the early months of 2018, but by 2019, ADA had prevailed and governed the area. A *denúncia* from January 2018 inside *Favela A* reads, "In [location], you can see close to 100 traffickers (unidentified), armed with automatic rifles and pistols, that invaded [location] last night and expelled the *milícias* that operated in the region." For the next several months, *denúncias* from inside the favela detail a brutal turf war between ADA and the *milícias*. Several months then pass with no *denúncias* about criminal groups until late 2019. In October 2019, one *denúncia* mentions the words "ADA FACTION" and "MILÍCIA," while several others just mention "ADA." A close reading of this first call summary reveals that ADA governs the community: "Inside the community... of the ADA faction... the traffickers walk around fully armed, with pistols and automatic rifles..." The mention of *milícias* is a red herring: "...[the traffickers] assassinated several people who were friends with *milícia* members..." Through a close read, the governing group is clear, but if using machine processing tools alone, one might misclassify this favela as still in conflict in 2019 because two groups are mentioned in the same call summary.

2. I made repeated attempts to better understand their coding methods in 2020-2022, and was met with the response that they could not be divulged. No further information has been published at the time of writing in 2026.

5. I construct four indicator variables (**CV, ADA, TCP, Milícia**) for the faction(s) present in each favela-election-year-month: October 2008, 2012, 2016, and 2020. I also construct an indicator variable if the favela is in a state of **Criminal conflict** at any point during the the six months prior to the election month. All of these indicators are used as controls for the main models, shown in Table 2. Finally, using the time series data, I calculate an indicator that takes a value of 1 if a favela experienced a change in criminal governance or a criminal conflict in the last 6 months, which I use in Section 5.1.

Table C2 then shows the distribution of favelas and favela population per elecion-year, by type of criminal presence. The favelas classified as "Criminal governance" are those for which there is evidence of one of the four governing factions governing that favela immediately before the municipal election for that year. The favelas classified as "Criminal conflict" are those for which there is evidence of an active criminal conflict or turf war at any point in the six months prior to the election. I denote those favelas where no faction was observed by *Disque Denúncia* as "No data."

Table C2: Favela distribution during election months, by type of criminal presence and by favela population

	2008		**2012**		**2016**		**2020**	
	Favelas	Pop.	Favelas	Pop.	Favelas	Pop.	Favelas	Pop.
Criminal governance	57%	84%	70%	88%	66%	88%	68%	89%
Criminal conflict	1%	2%	2%	4%	1%	2%	1%	1%
No data	42%	14%	28%	8%	33%	10%	31%	9%

There are two things to note in Table C2. First, while the election-year-to-election-year changes are greatest from 2008 to 2012, this can be explained by the relative sparsity of the data in 2008, since my dataset only begins in January 2008. Several of the favelas in the "No data" category in 2008 that are governed, but infrequently complained about, entered into the dataset for the remaining time periods. Second, while the proportion of governed favelas ranges from 57-70%, the proportion of governed residents ranges from the mid- to high-eighties range. The favelas with no data tend to be much smaller favelas, comprising fewer than 200 households. All large- and medium-sized favelas have a governing faction or are engaged in conflict for all four election-years.

C.2 Criminal governance proxy

I use the *Disque Denúncia* data to generate one other variable – criminal service provision – as a coarse approximation of criminal governing activity.[3] I classify calls with a binary

3. It is a coarse approximation because there is a broad range of behaviors that criminal governance can include (Arias 2017; Blattman et al. 2025; Lessing 2021; Magaloni et

indicator if they mention criminal service provision, especially "protection" or the provision of security. First, I assign *denúncias* a value of 1 if they mention one of three words about service provision: (i) *cobrança* (collection), (ii) *taxa* (fee/tax), or (iii) *segurança* (security, as in "protection"). Second, similar to the *criminal gatekeeping* and *criminal corralling* variables, I pass the *denúncias* through an additional "criminal service provision" filter and assign them a value of 1 if they mention a criminal organization (using the word lists in Table C1) *and* service provision.

Then, I restrict the sample of favelas to those where at least one call was registered that mentions criminal group service provision, and use this to subset the voting booth dataset. This sample is used for a robustness check (Table F5) that is described in Section 5.1.

C.3 Favela-level variables

1. **Distance to Favela**: This is the logged distance (in meters) from the voting booth to the boundary of the most-represented favela at the booth. The favela boundaries come from the 2016 favela shapefile published by the municipality,[4] and the polling station addresses from TRE-RJ. Polling stations are geocoded and distances calculated by the author.

2. **Population**: This is the logged population of the most-represented favela at the voting booth, produced by SABREN and available on the Rio de Janeiro municipality's open data portal.

3. **Income per capita**: This is the income per capita per favela from the 2010 Census, calculated directly by linking the favela shapefile to the the data from the CensoBR and geoBR data packages. I used the census tract-level data, aggregated to the favela level, to calculate average income per capita for each favela polygon.

C.4 Voting-booth-level variables

All voting-booth-level variables control for population dynamics inside the voting booth that might be correlated with favela residency. The historical process that generates the exogenous imbalance in voting booth composition is shaped by the different population dynamics in the favelas compared to the asphalt. For instance, sudden population growth in a favela may lead to the creation of a new voting booth with a high proportion of favela

al. 2020). Criminal service provision is one of the most common forms of criminal governance, but it can look different from place to place, sometimes resembling extortion and sometimes resembling taxation of other licit or illicit goods and services. I interpret the existence of criminal service provision as criminal governance, but the inverse does not necessarily hold: the absence of criminal service provision does not imply the absence of criminal governance. it may simply mean that the governing faction engages in other types of governing behaviors.

4. Available to download here: https://www.data.rio/datasets/PCRJ::limite-favelas-2016

residents. This ratio of favela to asphalt voters in the booth may persist over time, even as new asphalt and favela voters are being assigned to vote in that particular booth.

The *TSE* (Supreme Electoral Tribunal) publishes voting-booth-level statistics about the gender, age, education, and civil status of voters. I control for these time-variant demographic variables at the voting-booth-year-level in the main models:

1. **Age**: The *TSE* publishes the number of voters per ballot box of age 16, 17, and in binned categories for ages 18 and up. For all categories, I take the mean value, then calculate the weighted mean age for the entire ballot box.

2. **Education**: The *TSE* publishes the number of voters per education category, ranging from "illiterate" to "college complete." I calculate the mean education level per voting booth per year. The mean value for the main empirical strategy is an education level between 4 and 5, "primary school complete," and "high school started but incomplete," respectively.

3. **Percent women**: The *TSE* publishes the percent of voters per booth who identify as male and female. A negligible amount do not disclose or identify with the two above genders. I calculate the share of voters per ballot-box-year who identify as female.

4. **Percent married**: The *TSE* publishes the percent of voters in several civil status categories: married, divorced, single, widowed, and legally separated. Fewer than 5% of the sample is divorced, widowed, or legally separated; the vast majority is either single or married. I calculate and report the percent of voters per ballot-box-year who are married.

Figure C1 presents these four demographic variables at the voting booth level, as the exogenous variable, *Share favela*, increases. It shows that favela residents are slightly younger, slightly less likely to be married, and have less education.

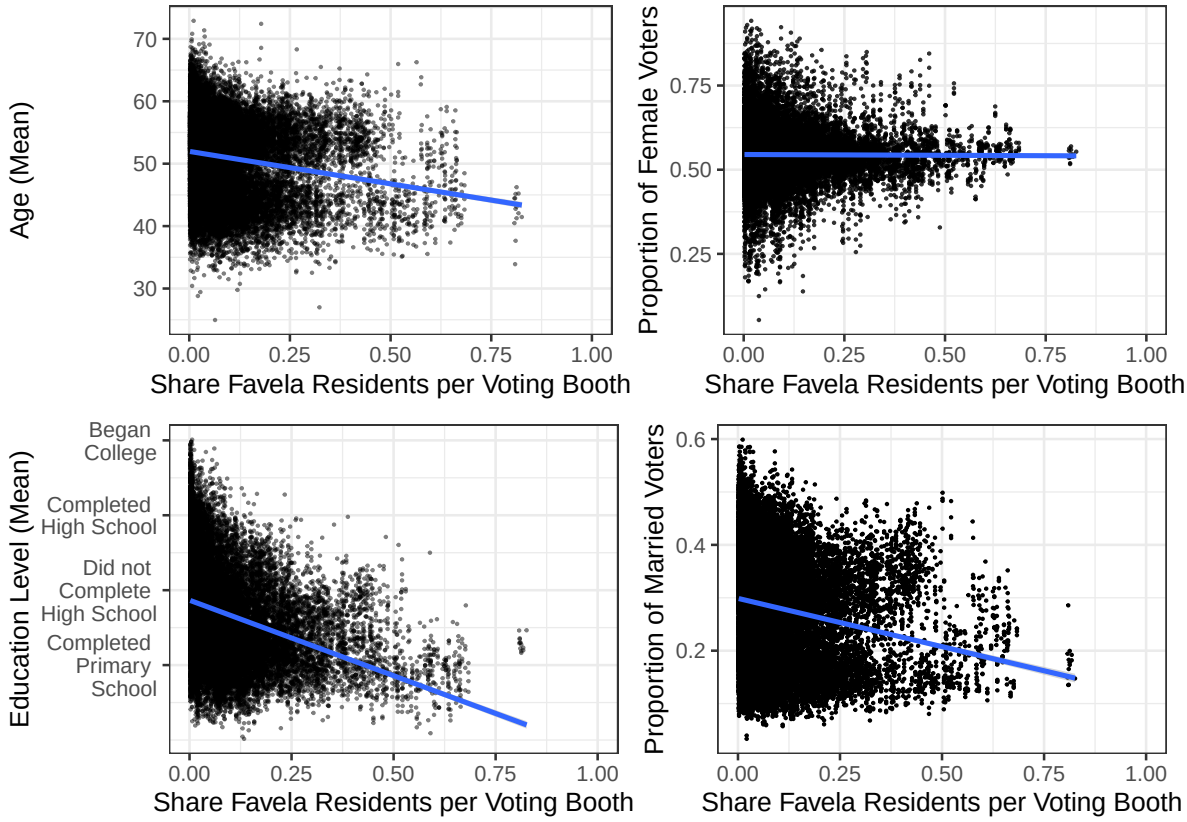
D Electoral assignment

This section describes the exogenous voting booth assignment procedure. As summarized in Section 4.2 of the main text, voters first choose their polling place within their electoral zone.[5] Polling places contain multiple booths, which contain approximately 300-500 voters.

Once voters choose a polling station (a primary school, etc.), the assignment process proceeds in the following way. Each year when new voters register, their information enters a federal system which assigns the voter to a booth *within their selected polling station*. Voting booth assignment stays constant year-to-year unless the voter changes their address or is removed from the list. The software prioritizes balance across voting booths, aiming for voting booths within the polling station to have an equal number of voters. When existing

5. In cities, electoral zones encompass thousands of people across several neighborhoods. They are territory-based.

Figure C1: Demographic characteristics of voting booths, by the share of favela residents



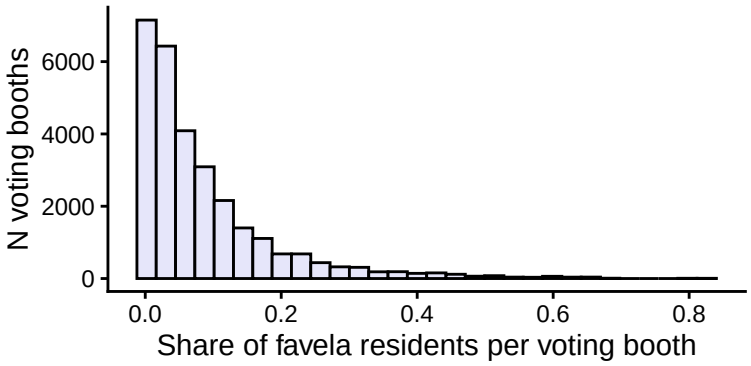
voting booths are approaching capacity, new ones are created. One common consequence of this process is that teenage new voters vote at different booths than their parents, despite living under the same roof.

I leverage this as-if random assignment to voting booths to calculate the share of favela residents that vote in each booth. I geocoded all postal codes within Rio de Janeiro.[6] Then, I filed an information request with the ombudsman's office of the Regional Electoral Tribunal of Rio de Janeiro (TRE-RJ) for data indicating the proportion of voters per voting booth that lived in postal codes located inside favelas.

All analyses in this paper focus on voting booths where favela residents vote. Figure D1 shows the distribution of the share of favela voters per booth. The mean voting booth in the sample has 8.6% of favela residents. The average range in $share_{max} - share_{min}$ within any one polling station is 5.7%, but there are many extreme examples. For instance, in the *Escola Municipal Tia Ciata* polling station, favela residents comprise less than 1% of voters in one booth, 19.8% in another, and a few classrooms down the hall, 63.5% of the voters at a different booth.

6. Brazil uses nine-digits postal codes, which are often as small as a city block in urban areas. Each favela contains many postal codes.

Figure D1: Share of favela residents per voting booth



The *Share favela* variable is important because it introduces exogenous variation, but also because it reduces measurement error in understanding how favela residents vote. Common polling station locations are schools or other government buildings that comply with accessibility and capacity codes. Often, such building only are located outside of favelas, or if inside a favela, are not large enough to accommodate all favela residents. Favela residents often vote in one of several close polling station to the favela boundary, where non-favela "asphalt" residents vote. Empirically, this is problematic for measuring how favela residents vote because one does not know which polling stations they travel to in order to cast a ballot or how integrated favela residents are with other voters.

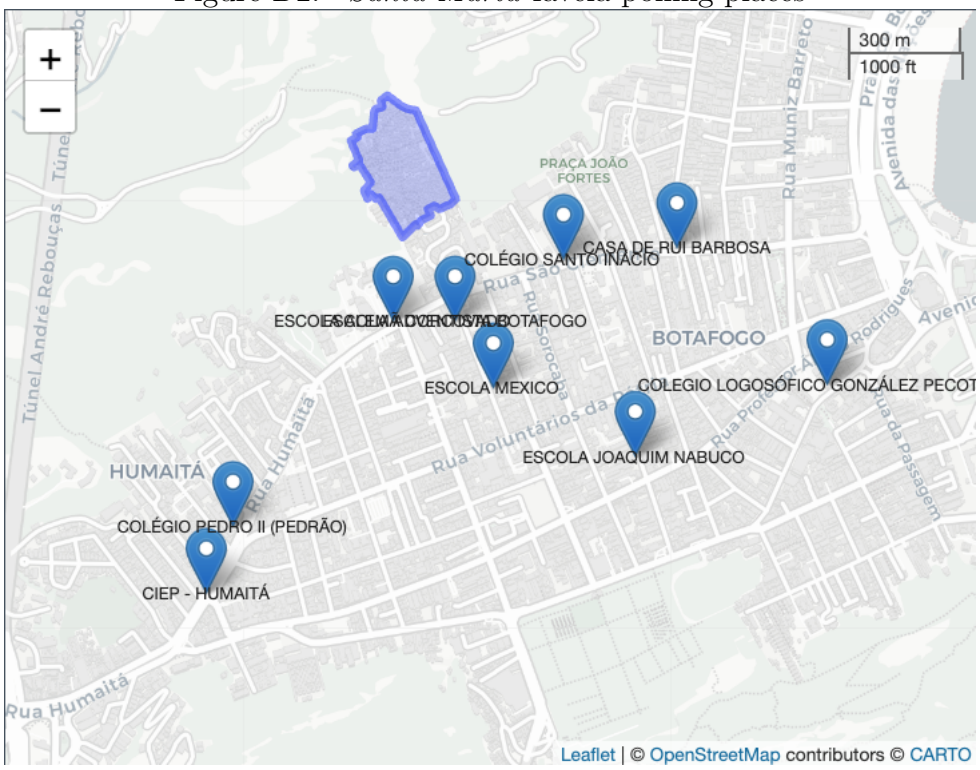
Figure D2 illustrates this problem in the case of the *Santa Marta* favela. There is no polling station inside *Santa Marta*, but there are several nearby. There is no public information about which polling station(s) *Santa Marta* residents are more likely to vote at versus which their asphalt neighbors, which is why all previous research attempting to map favelas onto voting outcomes have followed the same method: create an indicator variable for whether or not a polling station falls within a favela buffer zone, ranging from 250 to 1,000 meters outside the perimeter of the favela (Hidalgo and Lessing 2019; Nascimento 2017). In the example shown in Figure D1, all but three polling stations would take on a value of "1" for being within a 1,000 meter buffer zone of a favela.

My data reveals that there are many voting booths in this 1,000-meter range that only contain 5% or fewer favela voters. The primary school *Escola México*, is the polling place where a majority of *Santa Marta* residents vote, but is the fourth closest to the favela's main entrance. The composition of the seven voting booths inside *Escola México* range from 6-22% *Santa Marta* favela residents.

E Aggregate favela vote estimates

I analyze marginal candidate returns in the 2016 Rio de Janeiro City Council election as a test case, focusing on those who received between 6,023 and 16,064 votes, corresponding to the lowest-earning winner and the highest-earning loser, respectively. I estimate the number

Figure D2: *Santa Marta* favela polling places



Note: *Santa Marta* favela boundaries from IPP data, polling station locations from TRE-RJ, and basemap from OpenStreetMaps. *Escola México* is where a majority of *Santa Marta* residents vote. There are three closesr polling stations to *Santa Marta*, where fewer favela residents vote.

of favela votes a candidate receives in the following way:

1. **Lower bound:** I multiply *Share favela* by the total votes that a candidate received per booth. This assumes that favela and non-favela voters are uniformly likely to cast their ballot for any candidate, and is likely undercounting the votes that leading candidates win in voting booths with middling to high shares of favela residents.

2. **Upper bound:** I assume that all of a candidate's votes are from favela residents if they receive fewer votes than the total favela residents at the voting booth. For example, in a voting booth with 90 favela residents where the most-voted candidate receives 70 votes, I assume that all of the candidate's votes are coming from favela residents. This is an upper bound because it is likely attributing certain candidates' asphalt votes to favela votes.

3. **Mean:** I calculate the mean of the two above measures.

F Additional tables and figures

Additional tables and figures are shown below. Certain control coefficients are omitted for brevity, and all table notes indicate which of the main control variables or fixed effects are included. A full list of control and coefficient point estimates is available upon request.

Figure F1: Dependent variable distributions

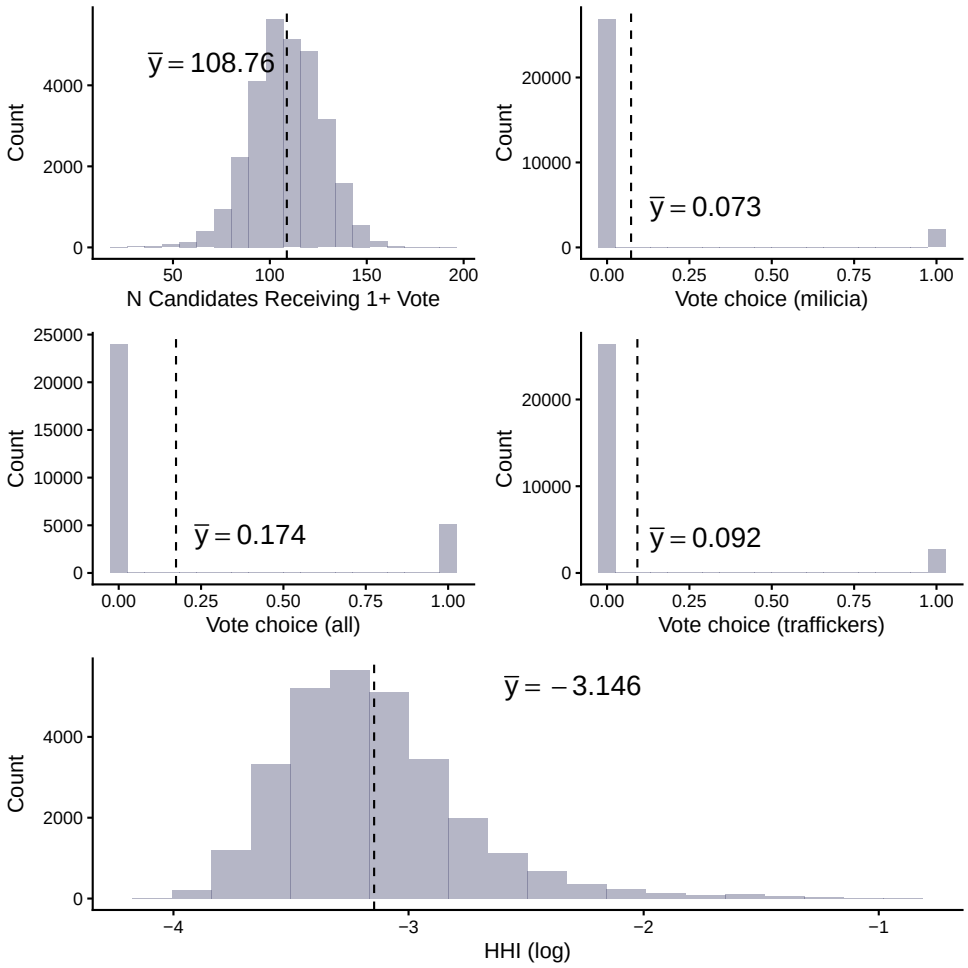


Table F1: Main models without control variables

	H1: Gatekeeping N candidates		**H2: Corralling** Vote choice (milícia)		Vote choice (all)	
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	−1.936**	−0.459**				
	(0.447)	(0.091)				
Criminal gatekeeping	−7.964**	−1.930**				
	(1.697)	(0.325)				
Criminal corralling × Share favela			0.068**	0.015**	0.071**	0.027**
			(0.021)	(0.005)	(0.017)	(0.006)
Criminal corralling			0.225**	0.036*	0.488**	0.135**
			(0.073)	(0.017)	(0.076)	(0.016)
Share favela	−1.346**	−1.341**	−0.006	−0.005	−0.006	−0.006
	(0.329)	(0.329)	(0.007)	(0.007)	(0.009)	(0.009)
Observations	29,026	29,026	29,026	29,026	29,026	29,026
R^2	0.551	0.552	0.516	0.515	0.489	0.488
	H3: Competition (DV: HHI)					
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	0.088**	0.018**			0.080**	0.003†
	(0.012)	(0.003)			(0.012)	(0.002)
Criminal gatekeeping	0.369**	0.078**			0.297**	0.006
	(0.053)	(0.011)			(0.050)	(0.007)
Criminal corralling × Share favela			0.114**	0.053**	0.084**	0.048**
			(0.016)	(0.010)	(0.018)	(0.011)
Criminal corralling			0.585**	0.238**	0.490**	0.229**
			(0.072)	(0.031)	(0.074)	(0.036)
Share favela	0.029**	0.030**	0.033**	0.033**	0.029**	0.032**
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Observations	29,026	29,026	29,026	29,026	29,026	29,026
R^2	0.673	0.675	0.677	0.689	0.680	0.689
Year FE	✓	✓	✓	✓	✓	✓
Station FE	✓	✓	✓	✓	✓	✓
Booth Controls						
Favela Controls						
Crim. Gov. Controls						

Note: Dependent variable (HHI) and *Share favela* variables are logged. Models include only polling station and yearly fixed effects. Standard errors are clustered at the polling station level: 1,338 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

Table F2: General election activity as an independent variable

	N candidates		Vote choice (milícia)		Vote choice (all)		HHI	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Election × Share favela	−0.723*	−0.234**	−0.015*	0.003**	−0.007	0.004**	0.043**	0.010**
	(0.321)	(0.031)	(0.007)	(0.001)	(0.012)	(0.001)	(0.009)	(0.001)
Election	−2.567*	−1.057**	−0.044	0.010**	0.075	0.027**	0.156**	0.046**
	(1.303)	(0.111)	(0.027)	(0.004)	(0.054)	(0.005)	(0.040)	(0.005)
Share favela	−1.659**	−1.600**	−0.003	−0.006	−0.007	−0.010	0.018**	0.016*
	(0.306)	(0.303)	(0.006)	(0.006)	(0.010)	(0.009)	(0.007)	(0.007)
Observations	29,026	29,026	29,026	29,026	29,026	29,026	29,026	29,026
R^2	0.570	0.574	0.516	0.517	0.485	0.491	0.677	0.691
IV	Binary	Count	Binary	Count	Binary	Count	Binary	Count

Note: Dependent variable (HHI) and *Share favela* variables are logged. Models include polling station and yearly fixed effects and demographic, favela, and criminal group controls. Standard errors are clustered at the polling station level: 1,338 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

Table F3: General gatekeeping or corralling activity as an independent variable

	H1: Gatekeeping	**H2: Corralling**		**H3: Competition**		
	N candidates	Milícia	All	HHI		
	(1)	(2)	(3)	(4)	(5)	(6)
Gatekeeping × Share favela	−1.392**			0.065**		0.041**
	(0.417)			(0.011)		(0.011)
Gatekeeping	−6.192**			0.281**		0.175**
	(1.566)			(0.045)		(0.046)
Corralling × Share favela		0.039*	0.063**		0.091**	0.073**
		(0.015)	(0.016)		(0.015)	(0.016)
Corralling		0.135*	0.413**		0.453**	0.382**
		(0.065)	(0.080)		(0.070)	(0.072)
Share favela	−1.632**	−0.006	−0.008	0.017**	0.021**	0.018**
	(0.304)	(0.006)	(0.009)	(0.007)	(0.007)	(0.007)
Observations	29,026	29,026	29,026	29,026	29,026	29,026
R^2	0.570	0.517	0.490	0.679	0.683	0.684

Note: Dependent variable (HHI) and *Share favela* variables are logged. Dependent variables listed as column headers. Models include polling station and yearly fixed effects and demographic, favela, and criminal group controls. Standard errors are clustered at the polling station level: 1,338 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

Table F4: Structural topic model indicators as independent variables

	N candidates			Vote choice (milícia)			Vote choice (all)			HHI		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
STM binary × Share favela	−0.905*			0.017			0.032*			0.043**		
	(0.351)			(0.012)			(0.015)			(0.012)		
STM binary	−4.939**			0.093†			0.229**			0.200**		
	(1.435)			(0.048)			(0.064)			(0.050)		
STM count × Share favela		−0.525**			0.009**			0.003			0.027**	
		(0.077)			(0.002)			(0.003)			(0.002)	
STM count		−3.030**			0.037**			0.063**			0.140**	
		(0.258)			(0.008)			(0.012)			(0.007)	
STM > 10 × Share favela			−0.447**			0.010**			0.004†			0.021**
			(0.077)			(0.003)			(0.003)			(0.002)
STM > 10			−2.409**			0.038**			0.059**			0.107**
			(0.262)			(0.009)			(0.010)			(0.008)
Share favela	−1.657**	−1.647**	−1.641**	−0.006	−0.006	−0.007	−0.010	−0.008	−0.009	0.018**	0.018**	0.018**
	(0.302)	(0.304)	(0.304)	(0.006)	(0.006)	(0.006)	(0.009)	(0.009)	(0.009)	(0.007)	(0.007)	(0.007)
Observations	29,026	29,026	29,026	29,026	29,026	29,026	29,026	29,026	29,026	29,026	29,026	29,026
R^2	0.571	0.575	0.574	0.518	0.518	0.519	0.490	0.491	0.492	0.679	0.698	0.692

Note: Dependent variable (HHI) and *Share favela* variables are logged. Models include polling station and yearly fixed effects and demographic, favela, and criminal group controls. Standard errors are clustered at the polling station level: 1,338 clusters. See Appendix A.2 for independent variable descriptions. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

Table F5: Main models with sample of criminally governed favelas

	H1: Gatekeeping		**H2: Corralling**			
	N candidates		Vote choice (milícia)		Vote choice (all)	
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	−2.097**	−0.415**				
	(0.606)	(0.110)				
Criminal gatekeeping	−7.963**	−1.718**				
	(2.267)	(0.366)				
Criminal corralling × Share favela			0.113**	0.019**	0.077**	0.031**
			(0.022)	(0.006)	(0.028)	(0.007)
Criminal corralling			0.338**	0.045*	0.525**	0.134**
			(0.069)	(0.018)	(0.078)	(0.020)
Share favela	−2.833**	−2.799**	0.022†	0.023†	0.022*	0.022*
	(0.641)	(0.642)	(0.012)	(0.012)	(0.009)	(0.009)
Observations	12,023	12,023	12,023	12,023	12,023	12,023
R^2	0.605	0.606	0.562	0.557	0.564	0.558
	H3: Competition (DV: HHI)					
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	0.095**	0.019**			0.083**	0.000
	(0.016)	(0.004)			(0.017)	(0.002)
Criminal gatekeeping	0.409**	0.078**			0.317**	−0.009
	(0.067)	(0.013)			(0.072)	(0.008)
Criminal corralling × Share favela			0.117**	0.056**	0.101**	0.056**
			(0.021)	(0.012)	(0.023)	(0.012)
Criminal corralling			0.633**	0.253**	0.566**	0.262**
			(0.067)	(0.035)	(0.072)	(0.036)
Share favela	0.031**	0.030**	0.035**	0.032**	0.029**	0.033**
	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)
Observations	12,023	12,023	12,023	12,023	12,023	12,023
R^2	0.717	0.721	0.728	0.752	0.733	0.752

Note: Dependent variable (HHI) and *Share favela* variables are logged. Models include polling station and yearly fixed effects and demographic, favela, and criminal group controls. Standard errors are clustered at the polling station level: 930 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

Table F6: Main models with alternate dependent variables

	N candidates with 5+ votes		Vote choice for *Milícia* candidates (post-2008) (*CPI* and DD)		(Just DD)		Most voted's share	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Criminal gatekeeping × Share favela	−0.232*	−0.073**					0.012**	0.000
	(0.099)	(0.016)					(0.003)	(0.000)
Criminal gatekeeping	−1.218**	−0.340**					0.053**	0.001
	(0.420)	(0.058)					(0.010)	(0.001)
Criminal corralling × Share favela			0.062**	0.010	0.004**	0.002**	0.014**	0.011**
			(0.020)	(0.008)	(0.001)	(0.001)	(0.004)	(0.002)
Criminal corralling			0.260**	0.038	0.017**	0.007**	0.081**	0.045**
			(0.079)	(0.027)	(0.006)	(0.002)	(0.015)	(0.008)
Share favela	−0.184**	−0.179**	−0.008	−0.008	−0.001	−0.001	0.002	0.002†
	(0.066)	(0.066)	(0.006)	(0.006)	(0.001)	(0.001)	(0.001)	(0.001)
Observations	29,026	29,026	23,260	23,260	23,260	23,260	29,026	29,026
R^2	0.535	0.537	0.515	0.513	0.539	0.539	0.656	0.665
IV	Binary	Count	Binary	Count	Binary	Count	Binary	Count
Timeframe	2008-20	2008-20	2012-20	2012-20	2012-20	2012-20	2008-20	2008-20

Note: Dependent variables are (1)-(2) = *N candidates with 5+ Votes*, (3)-(4) = Vote choice for *milícia*-linked candidates mentioned in either *Disque Denúncia* or the *CPI das Milícias*, (5)-(6) = Vote choice for *milícia*-linked candidates *only* mentioned in *Disque Denúncia*, (7)-(8) = *Most Voted's Share*. *Share favela* is logged and the sample is inclusive of only the 2012, 2016, and 2020 election years for columns (3)-(6). Models include polling station and yearly fixed effects and demographic, favela, and criminal group controls. Standard errors are clustered at the polling station level: 1,338 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

Table F7: Criminal gatekeeping and criminal corralling on voting, contested favelas

	H1: Gatekeeping		**H2: Corralling**			
	N candidates		Vote choice (milícia)		Vote choice (all)	
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × SF	−17.656**	−5.885**				
	(6.512)	(2.171)				
Criminal gatekeeping	−1664.439**	−554.813**				
	(70.133)	(23.404)				
Criminal corralling × SF			0.069	0.069	0.053	0.053
			(0.219)	(0.219)	(0.035)	(0.035)
Criminal corralling			0.222	0.222	−0.101	−0.101
			(0.248)	(0.248)	(0.310)	(0.310)
Share favela	3.147	3.147	−0.003	−0.003	−0.007	−0.007
	(1.905)	(1.906)	(0.006)	(0.006)	(0.012)	(0.012)
Observations	961	961	961	961	961	961
R^2	0.622	0.622	0.460	0.460	0.895	0.895
	H3: Competition (DV: HHI)					
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × SF	0.283†	0.094†			0.283†	0.094†
	(0.166)	(0.055)			(0.166)	(0.055)
Criminal gatekeeping	16.024**	5.341**			−15.405	−8.529
	(1.037)	(0.346)			(53 554.959)	(36 127.062)
Criminal corralling × SF			0.283†	0.283†		
			(0.166)	(0.166)		
Criminal corralling			−12.084**	−12.084**	−25.840	−34.211
			(0.506)	(0.506)	(44 030.564)	(89 106.169)
Share favela	−0.004	−0.004	−0.004	−0.004	−0.004	−0.004
	(0.018)	(0.018)	(0.018)	(0.018)	(0.018)	(0.018)
Observations	961	961	961	961	961	961
R^2	0.855	0.855	0.855	0.855	0.855	0.855

Note: Dependent variable (HHI) and *Share favela* (SF) variables are logged. Models include polling station and yearly fixed effects, and demographic and favela controls. Criminal gatekeeping and corralling are nearly collinear in this reduced sample, and after controlling for polling station and yearly intercepts, models with both interaction terms (**H3: Competition**, Models 5-6) drop the second interaction. Standard errors are clustered at the polling station level: 92 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

Table F8: Main Models Conditioning on Distance: $\leq$ 500m of Favela Perimeter

	H1: Gatekeeping	**H2: Corralling**		**H3: Competition**		
	N candidates	Milícia	All	HHI		
	(1)	(2)	(3)	(4)	(5)	(6)
Criminal gatekeeping × Share favela	−4.381**			0.160**		0.135**
	(1.272)			(0.033)		(0.036)
Criminal gatekeeping	−12.158**			0.500**		0.424**
	(3.020)			(0.079)		(0.083)
Criminal corralling × Share favela		0.230**	0.083†		0.157**	0.141**
		(0.043)	(0.048)		(0.046)	(0.052)
Criminal corralling		0.516**	0.474**		0.630**	0.571**
		(0.088)	(0.092)		(0.090)	(0.110)
Share favela	−3.586**	−0.012	−0.001	0.048**	0.055**	0.045**
	(0.835)	(0.008)	(0.006)	(0.017)	(0.019)	(0.018)
Observations	9,505	9,505	9,505	9,505	9,505	9,505
R^2	0.568	0.533	0.534	0.661	0.666	0.672

Note: Dependent variable (HHI) and *Share favela* are logged. Models include polling station and yearly fixed effects, and demographic, favela, and criminal group controls. Standard errors are clustered at the polling station level: 523 clusters. $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$.

G Research ethics

I confirm that the human subjects research pursued complies with the *Principles and Guidance for Human Subjects Research*. The human subjects evidence presented in this paper includes the semi-structured interviews with candidates, staffers, community leaders, and favela residents. The research was approved and designated exempt by the Committee on the Use of Human Subjects (CUHS) at Harvard University under Protocol number IRB18-1305.

Interviews. All interviews were conducted in person by the author in Brazilian Portuguese. Verbal consent was sought and obtained before the interview began, and there was no deception involved. To further protect confidentiality, interviews were not audio recorded, the author just took written notes. All identities are anonymized, and translated text is lightly edited for clarity. Respondents were informed of this and were reminded that they did not have to answer all questions. List of cited interviews:

- Interview 1: NGO leader, August 2018.
- Interview 2: Candidate 1, December 2019.
- Interview 3: Candidate 2, February 2021.
- Interview 4: Staffer 1, August 2018.
- Interview 5: Candidate 3, November 2018.
- Interview 6: Public Security Professional, October 2023.
- Interview 7: Campaign manager 1, September 2018.

Voter complaints. An additional data source used in this project is drawn from the *Disque Denúncia* anonymous complaints. While the *Disque Denúncia* data is used to create variables used for quantitative analysis, I also include the text of select complaints as qualitative evidence of the gatekeeping and corralling concepts. This qualitative data source is not human subjects data, as all complaints are anonymous and there is no personally identifying information that could link the complainant to a specific person. Despite this, the complaints contain sensitive information and should be treated with care. All complaints contain the name of rule-breaking candidates and neighborhoods where the offense was committed. To protect residents of these communities from retaliation at the group-level, I have removed all candidate- and neighborhood-level details from the complaints, as well as any other potentially identifying data. Complaints printed in the paper's text contain the translated text with *ALL-CAPS* placeholders for generic candidate or location names. Further, I remove all other potentially identifying details from the complaints and only report the month and year of the tip. List of cited complaints:

- Complaint 1: October 2020.
- Complaint 2: November 2020.
- Complaint 3: November 2020.
- Complaint 4: October 2012.
- Complaint 5: October 2016.

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